SI Appendix:
Training-Free Generation of Protein Sequences from Small Family Alignments via Stochastic Attention

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S1 Derivation of the Stochastic Attention Model

Sampling from a broad class of generative diffusion models reduces to one task: draw a state vector $\boldsymbol{\xi} \in \mathbb{R}^d$ from a target density p written in Gibbs (Boltzmann) form as:

$$p(\boldsymbol{\xi}) \;\propto\; \exp\bigl(-U(\boldsymbol{\xi})\bigr),$$

where $U(\boldsymbol{\xi}) \in \mathbb{R}$ is the *potential* (negative log-density) of configuration $\boldsymbol{\xi}$ (low U = high probability). The *Unadjusted Langevin Algorithm* (ULA) [1] follows the potential gradient downhill while injecting noise, with step size $\tilde{\alpha} > 0$:

$$\boldsymbol{\xi}_{t+1} \;=\; \boldsymbol{\xi}_t \underbrace{-\,\tilde{\alpha}\,\nabla U(\boldsymbol{\xi}_t)}_{\text{score step}} + \sqrt{2\tilde{\alpha}}\,\boldsymbol{\epsilon}_t, \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d).$$

The ULA requires a *score function* $\nabla \log p = -\nabla U$. In diffusion and score-based models the score is *typically not known* in closed form, so a neural network must *learn* it from a large dataset of samples. Suppose we are interested in K stored patterns $\mathbf{m}_k \in \mathbb{R}^d$, gathered as the columns of a *unit-normalized* memory matrix: $\mathbf{X} = [\mathbf{m}_1, \dots, \mathbf{m}_K] \in \mathbb{R}^{d\times K}$. The *modern Hopfield log-sum-exp (LSE) energy* [2] (with inverse temperature $\beta > 0$) defines the landscape:

$$E(\boldsymbol{\xi}) \;=\; \tfrac{1}{2}\,\|\boldsymbol{\xi}\|_2^2 \;-\; \tfrac{1}{\beta}\,\log\left(\sum_{k=1}^{K} \exp\bigl(\beta\,\mathbf{m}_k^\top \boldsymbol{\xi}\bigr)\right),$$

whose Boltzmann distribution $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta E(\boldsymbol{\xi}))$ is our target. For this target the score need not be learned. The gradient of E is *exactly* the attention residual (the current state minus the attention output), so:

$$\underbrace{\nabla \log p_\beta(\boldsymbol{\xi})}_{\text{score}} \;=\; -\beta\,\nabla E(\boldsymbol{\xi}) \;=\; -\beta\Bigl\{\boldsymbol{\xi} \;-\; \underbrace{\mathbf{X}\,\mathrm{softmax}\bigl(\beta\,\mathbf{X}^\top\boldsymbol{\xi}\bigr)}_{\mathbf{T}(\boldsymbol{\xi})}\Bigr\}.$$

Here $\mathrm{softmax}$ turns the pattern similarities $\beta\,\mathbf{X}^\top\boldsymbol{\xi}$ into nonnegative weights that sum to one, so $\mathbf{T}(\boldsymbol{\xi})$ is just a weighted average of the stored patterns: one attention call returns the exact score. Substitute the closed-form score into ULA with potential $U = \beta E$ and step size $\tilde{\alpha} = \alpha/\beta$, where the mixing coefficient $\alpha \in (0,1)$ sets how far each step moves. The score step $-\tilde{\alpha}\nabla U = \alpha\bigl(\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi}\bigr)$ gives the update:

$$\boxed{\boldsymbol{\xi}_{t+1} \;=\; \underbrace{(1-\alpha)\,\boldsymbol{\xi}_t}_{\text{contraction}} + \underbrace{\alpha\,\mathbf{X}\,\mathrm{softmax}\bigl(\beta\,\mathbf{X}^\top\boldsymbol{\xi}_t\bigr)}_{\text{attention pull to memories}} + \underbrace{\sqrt{\tfrac{2\alpha}{\beta}}\,\boldsymbol{\epsilon}_t}_{\text{Gaussian noise}}}$$

S2 Amino Acid Composition Profiles

We compared the per-position amino acid frequency profiles of the stored seed sequences and stochastic attention (SA)-generated sequences for the RNA recognition motif (RRM) family (PF00076, Fig. S1). The generated ensemble closely reproduces the conservation pattern of the family: strongly conserved positions (e.g., the aromatic residues characteristic of the ribonucleoprotein motifs RNP1 and RNP2) retain high frequency in the generated output, while variable positions display a similar breadth of amino acid usage. This agreement is consistent with the low Kullback–Leibler (KL) divergences reported in Table S1 and indicates that the Boltzmann sampling procedure captures not only global amino acid composition but also the position-specific constraints encoded in the seed alignment.

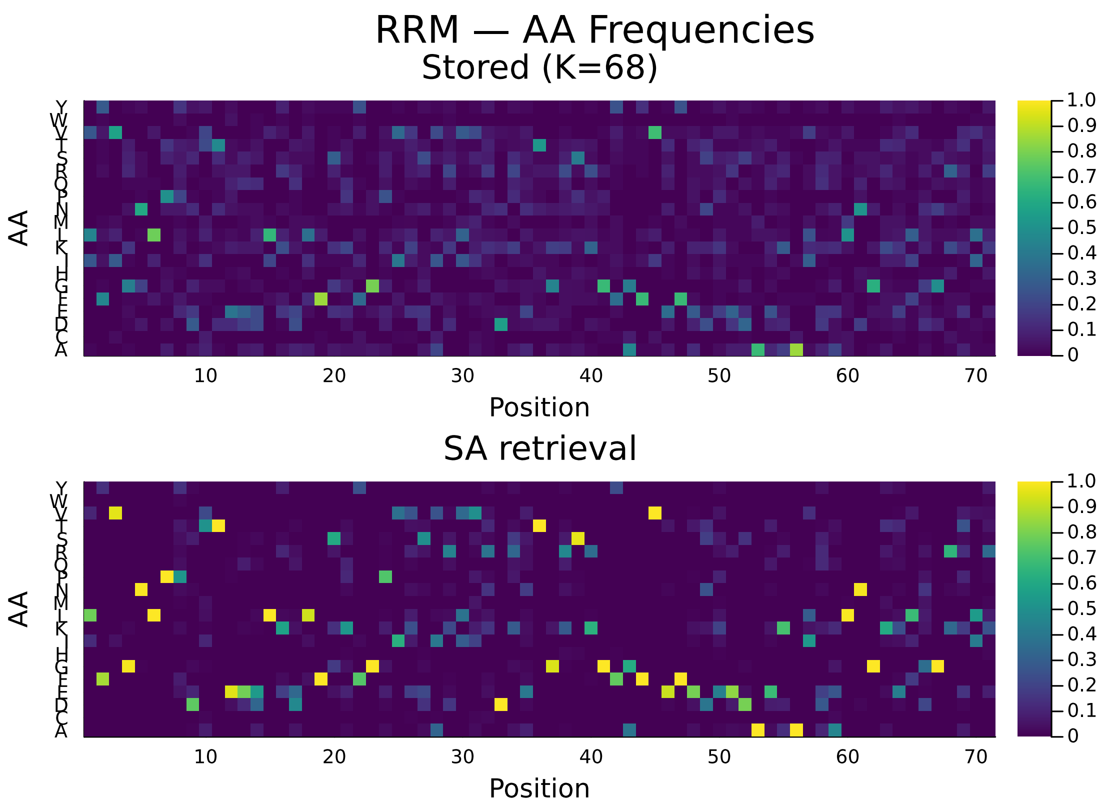

Figure S1: Per-position amino acid frequency profiles for the RRM family (PF00076). **Top:** stored seed sequences ($K{=}68$). **Bottom:** SA-generated sequences ($S{=}150$, retrieval regime). The generated ensemble reproduces the position-specific conservation pattern of the family, including strongly conserved positions and variable regions.

S3 Generation Metrics

Table S1 reports the full generation metrics for all eight Pfam families across five sampling conditions: SA generation ($\beta \approx 2\beta^*$), SA retrieval ($\beta \approx 20\beta^*$), bootstrap resampling, Gaussian perturbation, and convex combination. For each condition, we report the KL divergence of amino acid composition, principal component analysis (PCA)-space cosine novelty, and nearest sequence identity to a stored pattern. Standard errors (SE) are computed over 30 independent Langevin chains of 5 samples each; KL standard errors are obtained via 1,000-fold bootstrap resampling. Bold entries indicate the best-performing method for each family and metric. The generation regime provides the best balance of compositional fidelity, novelty, and diversity across all eight families, while the convex combination baseline substantially increases KL divergence in most families due to interpolation artifacts in PCA space.

Table S2 reports the per-family generation metrics for the three learned baselines, profile HMM emit, EvoDiff, and the MSA Transformer, that stochastic attention is compared against in the main-text head-to-head analysis. For each family and method, the amino acid composition KL divergence, the PCA-space cosine novelty, and the nearest sequence identity to a stored pattern were computed under the same 150-sequence budget and evaluation pipeline applied to SA generation, and are reported as the mean and standard error over 30 sub-blocks of the generated set, following the convention of Table S10. The learned baselines attained low KL divergence in most families, but their nearest sequence identities fell below the empirical within-family nearest-neighbor identity envelope (Table S14) in several families, with profile HMM emit reaching 11% and the MSA Transformer 9% in Pkinase, below the minimum nearest-neighbor identity of any natural sequence in that family (28%), indicating compositional drift away from the stored patterns rather than family-consistent generation. EvoDiff showed elevated composition divergence on the longest family (Pkinase, KL$=0.173$, more than an order of magnitude above its values in the other seven families); this largely reflects EvoDiff generating sequences well short of the alignment length (≈ 199 residues vs. $L=262$), which the length-normalization step pads with alanine, inflating the apparent alanine frequency, rather than an intrinsic compositional failure of the model. Together with the SA generation metrics in Table S1 and the Potts comparison in Table S10, these values constitute the complete per-family record underlying the baseline comparison.

S4 Scaling Study and Multi-Family Generalization

To characterize how generation quality degrades with decreasing family size, we subsampled the WW domain ($K=420$) at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates per condition (Fig. S2). In the generation regime, stochastic attention maintained low KL divergence across the entire range (0.019 ± 0.008 at $K=20$, improving to 0.010 ± 0.002 at $K=400$), while the convex combination baseline exceeded $\mathrm{KL}=0.8$ at every family size. Novelty increased with K (from 0.47 to 0.74), reflecting the richer landscape of a larger memory set, while nearest sequence identity decreased correspondingly (from 0.71 to 0.56). The empirical critical temperature β^* increased from 2.7 at $K=20$ to 5.5 at $K=400$, tracking the growth in PCA dimensionality. These results establish that SA generation is robust down to alignments as small as 20 sequences, though with reduced novelty reflecting the more constrained energy landscape.

The WW-only scaling study showed robustness down to $K=20$, but a single family cannot establish generality. To test whether this robustness holds across families with different sequence lengths, conservation levels, and PCA dimensionalities, we subsampled three additional families to $K=20$ with five independent replicates each: SH3 ($K_{\mathrm{full}}=55$, $L=48$), Kunitz ($K_{\mathrm{full}}=99$, $L=53$), and zf-C2H2 ($K_{\mathrm{full}}=151$, $L=23$). All stochastic attention parameters were identical to the main experiment; PCA and β^* were recomputed from scratch for each subsample. Table S3 reports the results alongside the original WW subsamples at $K=20$. Stochastic attention maintained low KL divergence (0.013–0.026), substantial novelty (0.46–0.47), and moderate sequence identity (0.67–0.73) across all four families, with narrow standard deviations across the five replicates. The consistency of these metrics across families that differ in length by more than a factor of two (zf-C2H2: $L=23$ vs. Kunitz: $L=53$) indicates that the entropy inflection criterion adapts the operating temperature to each family's geometry, and that the robustness observed in the WW scaling study is not family-specific.

S5 AlphaFold2 Cross-Validation

Figure S3 shows predicted local distance difference test (pLDDT) and template modeling (TM)-score results from both ESMFold and AlphaFold2 side by side across all eight families. The two predictors produce similar qualitative patterns: sequences generated by stochastic attention achieve pLDDT well above the 70 threshold in every family under both predictors, and the family-to-family ordering of TM-scores is preserved. Template modeling scores were significantly higher for SA generation than for stored sequences in all eight families ($p < 0.05$; Cohen's d=0.47–1.73). Figure S4 quantifies the agreement directly: TM-scores are strongly concordant across predictors (Pearson r=0.999), supporting the conclusion that the structural ranking of conditions is not specific to ESMFold's architecture or training data. The pLDDT concordance is more moderate (r=0.434), reflecting the fact that each predictor has its own confidence calibration scale; AlphaFold2 pLDDT values are systematically higher

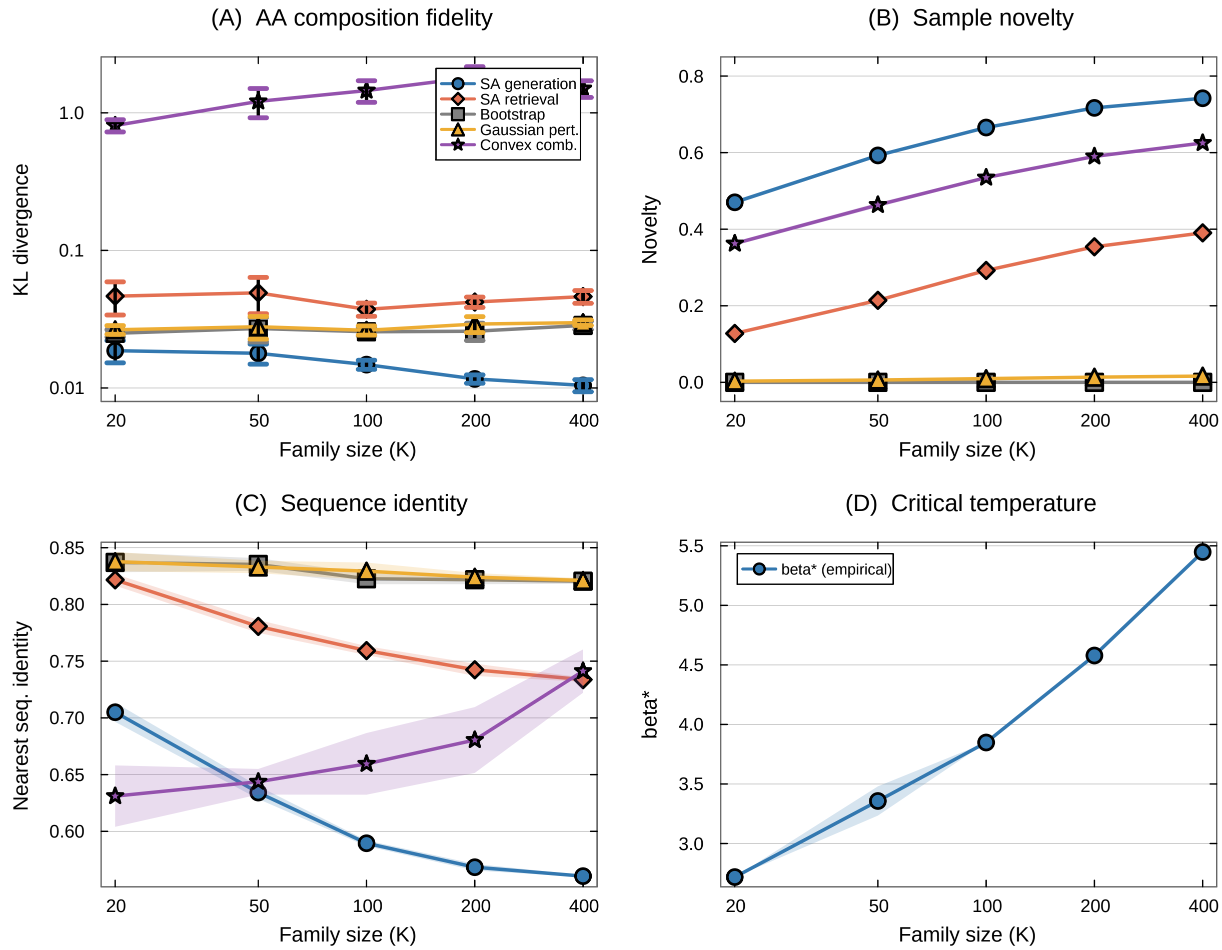

Figure S2: Scaling study: WW domain subsampled at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates. **(A)** KL divergence (log–log scale). Stochastic attention generation maintains low KL across the entire range; convex combination substantially increases KL. **(B)** Novelty increases with K for SA methods; baselines remain near zero. **(C)** Nearest sequence identity decreases with K, indicating the sampler explores further from stored patterns as more patterns shape the energy landscape. **(D)** The empirical critical temperature β^* increases with family size. Shaded regions and error bars show ± 1 SE.

than ESMFold values for the same sequences, consistent with known differences in the two models' confidence scoring. The strong TM-score concordance supports the structural conclusions drawn from ESMFold in the main text.

Reference structures

Table S4 lists the experimentally determined structures used as TM-align references for each family. References are canonical, high-resolution structures of the cognate domain.

S6 Consensus Proximity and Predicted Structural Quality

Sequences generated by stochastic attention achieved template modeling (TM)-scores that, in several families, exceeded those of the stored natural sequences. Because all structure predictors are trained on evolutionary data, one explanation is predictor bias: a predictor might assign higher confidence to sequences that lie closer

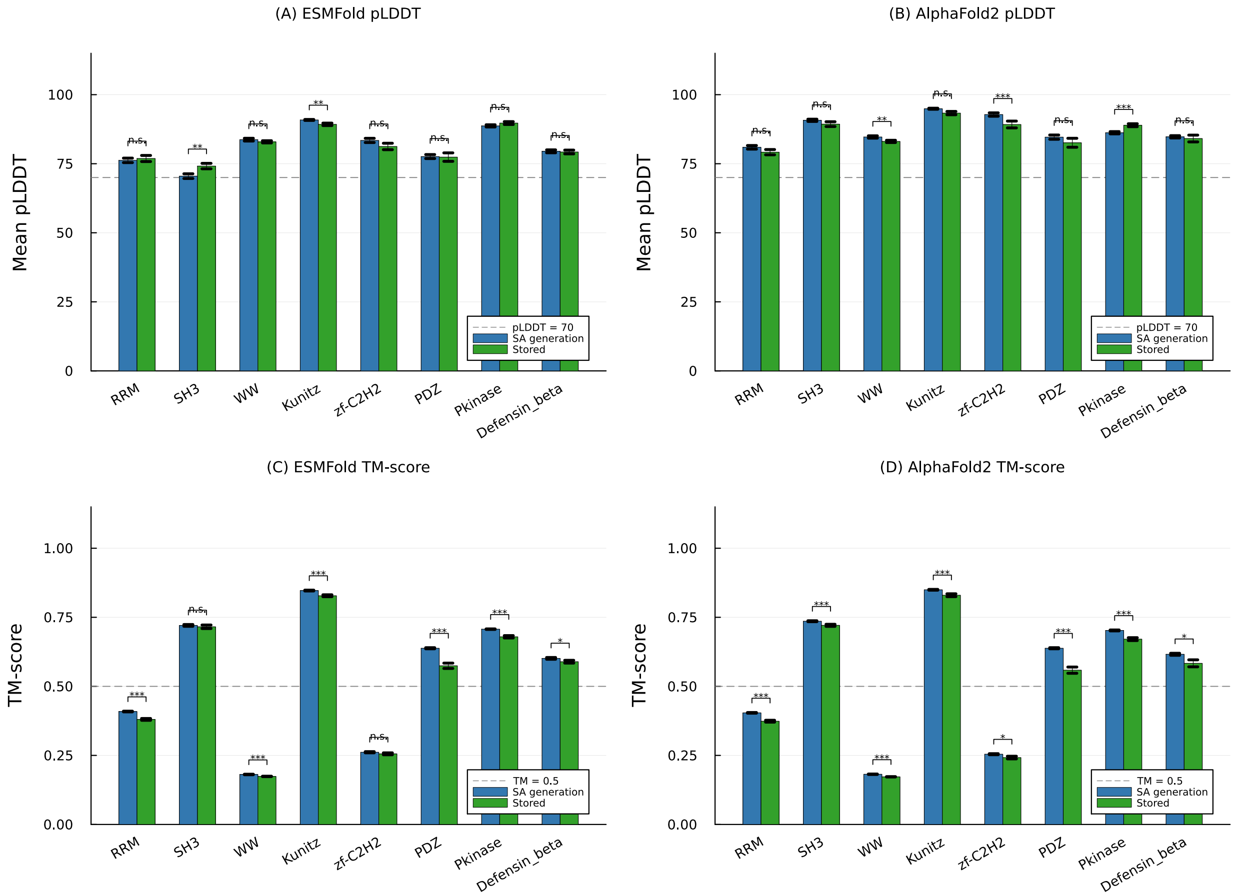

Figure S3: Structure validation results from ESMFold **(A, C)** and AlphaFold2 **(B, D)** across all eight Pfam families. **(A, B)** Mean pLDDT for SA-generated (blue) and stored (green) sequences; dashed line at pLDDT$=70$. **(C, D)** Mean TM-score to the family reference structure; dashed line at TM$=0.5$. Significance brackets show Wilcoxon rank-sum tests comparing SA generation to stored sequences. The two predictors produce similar qualitative patterns across all families.

to the family consensus, in which case the elevated TM-scores would be an artifact of consensus proximity rather than evidence of structural quality. To test this directly, we asked whether predicted structural quality tracks consensus proximity *within* each family. For every SA-generated sequence whose structure was predicted (n=50 per family), we computed its identity to the family consensus (the most frequent residue per column among stored sequences) and its mean predicted local distance difference test (pLDDT) score and reference-normalized TM-score, then computed the within-family Spearman rank correlation between consensus identity and each structural metric (Table S5).

The association is weak and inconsistent in sign. Across families, the Spearman correlation between consensus identity and TM-score ranged from -0.13 to 0.53 (negative in SH3, WW, and Pkinase), and between consensus identity and pLDDT from -0.10 to 0.43. Pooling the within-family-standardized values gave ρ=0.17 for TM-score (p=5×10^{-4}, n=400) and ρ=0.23 for pLDDT (p=2×10^{-6}); each accounts for under 4% and 6% of the variance, respectively. Consensus distance is therefore not the dominant determinant of predicted structural quality, which argues against the interpretation that the elevated TM-scores are primarily an artifact of consensus proximity. This analysis does not exclude an *absolute* preference, shared by both the sampler and the predictor, for family-typical sequences; a complementary control that folds the consensus-with-noise ensemble (Section S17) through the same predictors would further isolate that effect, and definitive resolution requires experimental structure determination. We also note that SA generation is most consensus-proximal in the Pkinase family (mean identity to consensus 0.73, the highest of any family), which accounts for both its near-natural

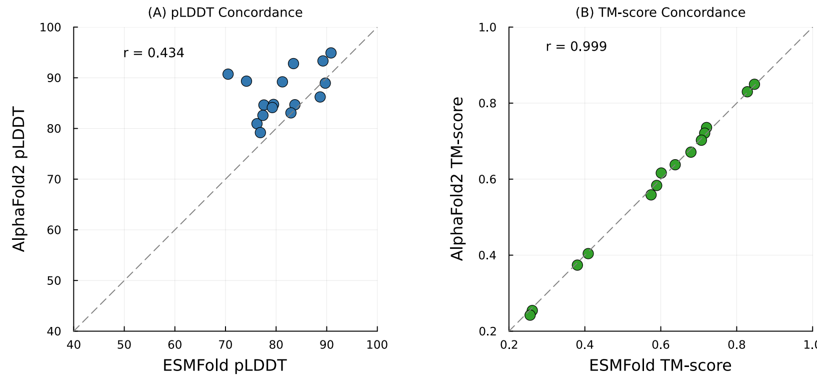

Figure S4: Concordance between ESMFold and AlphaFold2 structure predictions for SA-generated sequences across all eight families. Each point is one family-condition pair. **(A)** pLDDT concordance (r=0.434); AlphaFold2 pLDDT values are systematically higher, reflecting differences in confidence calibration. **(B)** TM-score concordance (r=0.999); points lie near the identity line, indicating strong agreement in condition ranking.

nearest-neighbor identity and its low mutual-information correlation (r=0.05; Section S8), since near-consensus sequences carry little pairwise covariation.

S7 Sequence-Level Analysis

We aligned the five highest-confidence SA-generated Kunitz domain sequences against the family consensus to examine how the method handles the interplay between conservation and variation at the single-residue level (Fig. S5). All five sequences preserved every highly conserved position (11 of 11 positions with $> 90\%$ conservation in the seed alignment) and all six cysteines that form the three disulfide bonds essential to the Kunitz fold, while introducing 17–25 substitutions concentrated at variable positions. Per-position Shannon entropy correlated strongly between the SA-generated ensemble and the stored multiple sequence alignment (MSA; r=0.968), indicating that the method distinguishes conserved positions from positions that tolerate variation. The concentration of substitutions at high-entropy positions is consistent with the Boltzmann sampling mechanism: the softmax attention weights distribute probability mass across stored patterns in proportion to their similarity to the current sample, so strongly conserved positions (where all stored patterns agree) are decoded to the consensus residue, while variable positions (where stored patterns diverge) sample from the local distribution of amino acids.

S8 Pairwise Mutual Information Analysis

To assess whether SA-generated sequences preserve pairwise residue covariation, we computed mutual information (MI) matrices for the stored seed alignment and for sequences generated by each method. For each family, we constructed the $L \times L$ MI matrix, where position pair (i, j) has MI defined as $\mathrm{MI}(i, j) = \sum_{a,b} p(a, b) \log \frac{p(a,b)}{p_i(a)\, p_j(b)}$, with an additive pseudocount of 1.0 to regularize sparse counts. We then extracted the upper-triangle elements ($L(L-1)/2$ pairs) and computed Pearson and Spearman correlations between the stored MI vector and the MI vector of each generated ensemble. Table S6 reports the results across all eight families.

The central observation is that SA-generated sequences preserve much of the pairwise covariation structure of the stored alignment: Pearson r=0.53–0.92 in seven of eight families (mean r=0.72 excluding Pkinase), with SA outperforming profile hidden Markov model (HMM) sequences in seven of eight families (mean r=0.42), which model each position independently. The Pkinase family (K=37, L=262) was an outlier, consistent with the high data-to-position ratio limiting MI estimation quality in both the stored and generated ensembles. Methods that explicitly model pairwise couplings (Potts/plmDCA: r=0.65–0.96 in seven of eight families, with Defensin_beta an outlier at r=0.23) or leverage pretraining on millions of MSAs (EvoDiff: r=0.59–0.97) generally achieved

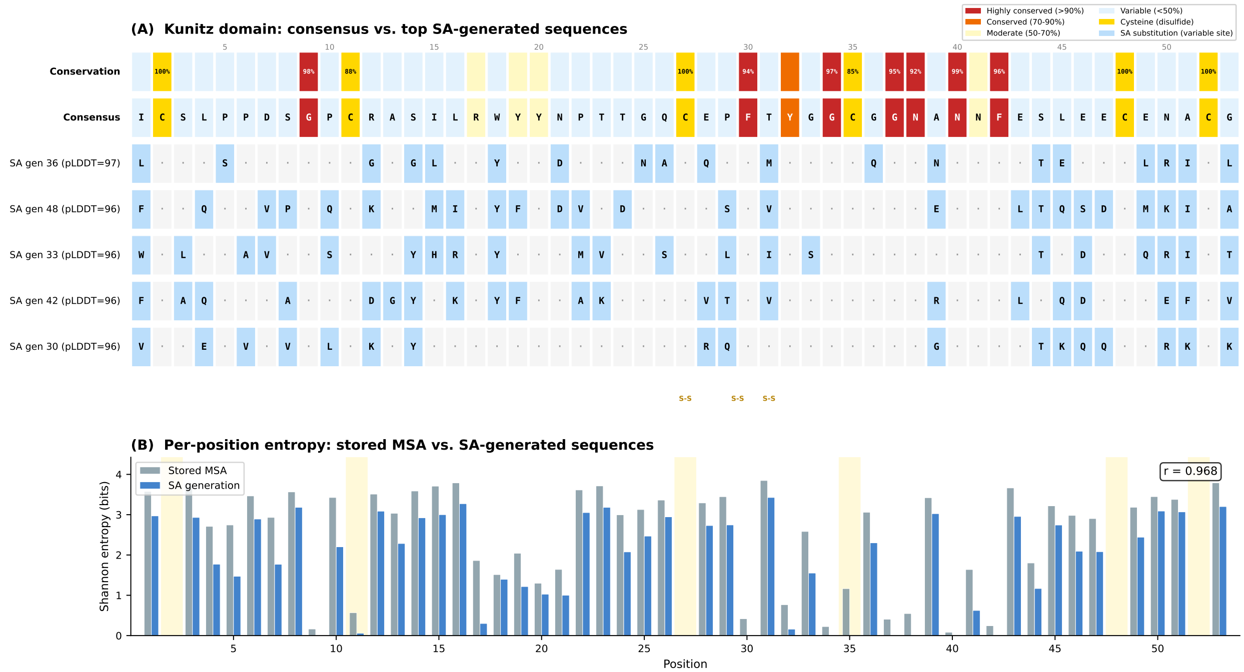

Figure S5: Sequence-level analysis of SA-generated Kunitz domain sequences. **(A)** Alignment colored by conservation level. **(B)** Per-position Shannon entropy correlation between the SA-generated ensemble and the stored seed alignment (r=0.968).

higher MI correlations, as expected given their direct access to pairwise or higher-order statistics. The ordering HMM $<$ SA $<$ Potts $\leq$ EvoDiff held in three of eight families and is consistent with the information each method encodes: position-independent frequencies, the implicit all-order couplings of the Hopfield energy, explicitly fitted pairwise couplings, and deep coevolutionary priors from large-scale pretraining, respectively. The Defensin_beta exception (Potts r=0.23, below both SA and HMM) reflects the same compositional limitation observed in the generation metrics (Table S10). Figure S7 shows this ordering across all eight families simultaneously, and Fig. S6 shows MI scatter plots for representative families, with structural contact pairs highlighted to assess whether SA preserves covariation at structurally relevant positions.

S9 Biophysical Properties of Beta Defensin Sequences

Figure S8 shows the biophysical properties of SA-generated beta defensin sequences (PF00711) compared to stored natural sequences and baseline methods. Sequences generated by stochastic attention preserved the six-cysteine disulfide scaffold, maintained a cationic charge profile, and occupied the same region of charge-hydrophobicity space as natural defensins.

S10 Independent Fitness Validation via ESM2

As an independent assessment of whether SA-generated sequences resemble natural proteins, we scored all generated and stored sequences using ESM2-650M [3], a protein language model pretrained on millions of UniRef50 sequences that is entirely independent of the SA framework. For each sequence, we computed the pseudo-perplexity via masked marginal scoring: each position was masked in turn, the model predicted a distribution over amino acids, and the log-probability of the true residue was recorded. The pseudo-perplexity (lower is more protein-like) is $\exp(-\frac{1}{L}\sum_{\ell=1}^{L}\log P(a_\ell \mid \mathbf{a}_{\setminus\ell}))$, where a_ℓ is the amino acid at position ℓ and $\mathbf{a}_{\setminus\ell}$ denotes all other positions. Wilcoxon rank-sum tests with Cohen's d effect sizes compared SA-generated sequences

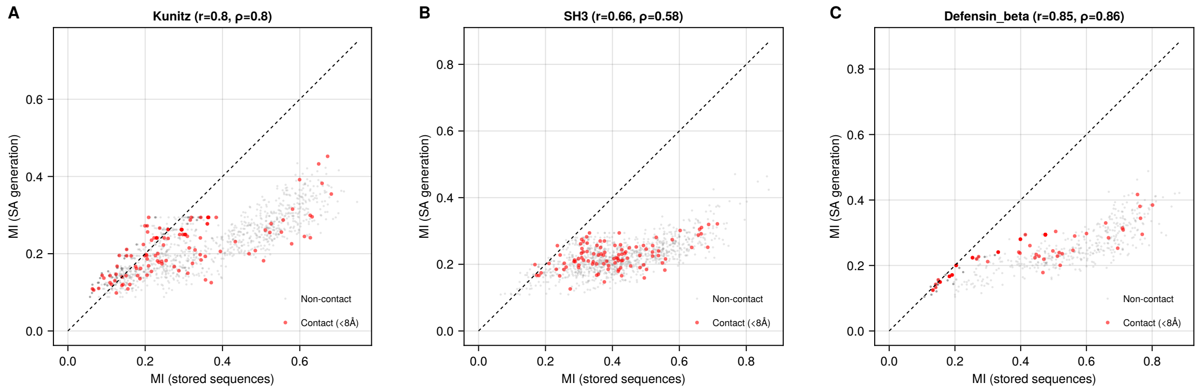

Figure S6: Pairwise MI scatter plots for representative families. Each point represents one position pair (i, j); the x-axis is MI from the stored seed alignment and the y-axis is MI from SA-generated sequences. Red points indicate structurally contacting residue pairs (Cα–Cα distance <8 Å, sequence separation ≥ 5 residues). Pearson r and Spearman ρ are shown in each panel.

(n=150) to stored natural sequences (n=37–420) for each family (Table S7).

Sequences generated by stochastic attention were statistically indistinguishable from natural family members in four of eight families (Wilcoxon $p > 0.05$; RRM, zf-C2H2, Pkinase, Defensin_beta: $p > 0.29$). In the Kunitz domain, SA generation achieved significantly *lower* (better) pseudo-perplexity than stored sequences (4.47 vs. 4.98, $p < 0.001$, d=-0.74). Three families showed modestly higher pseudo-perplexity for SA generation (SH3: d=0.60; WW: d=1.03; PDZ: d=0.60), though all SA-generated sequences remained well within the range of natural protein sequences. The overall mean pseudo-perplexity across all families was 5.71 for SA-generated sequences versus 5.58 for stored sequences, indicating that an independent protein language model trained on billions of sequences scores SA-generated sequences comparably to natural family members.

S11 Deep Mutational Scanning Validation

To assess whether SA-generated substitutions correspond to experimentally tolerated mutations, we cross-referenced the amino acid changes introduced by SA generation against published deep mutational scanning (DMS) datasets for two families: the PDZ domain (DLG4/PSD-95 PDZ3; McLaughlin et al. [4]) and the SH3 domain (GRB2 SH3; Faure et al. [5]). For each family, we identified the DMS reference protein in the Pfam seed alignment, mapped alignment positions to DMS coordinates, and classified each SA-generated substitution relative to the family consensus as experimentally tolerated ($\mathrm{DMS\_score\_bin}$=1) or deleterious ($\mathrm{DMS\_score\_bin}$=0) based on the published fitness data (Table S8).

Substitutions generated by stochastic attention were enriched for experimentally tolerated mutations in both families. In the PDZ domain, 91.7% of the $4{,}313$ matched substitutions were classified as tolerated, compared to an overall tolerance rate of 77.5% in the DMS dataset ($1.18\times$ enrichment, binomial $p < 10^{-134}$). The continuous DMS fitness scores of SA-generated substitutions were also significantly higher than those of random single mutations (Wilcoxon rank-sum $p < 10^{-20}$, Cohen's d=0.42). In the SH3 domain, the enrichment was larger: 90.4% of $2{,}396$ matched substitutions were tolerated versus 67.5% overall ($1.34\times$ enrichment, binomial $p < 10^{-155}$), with a larger effect size on continuous fitness scores (d=0.70, $p < 10^{-54}$). These results indicate that the Boltzmann sampling procedure preferentially generates substitutions at positions and to residues that are individually tolerated in single-mutant assays, without any explicit fitness objective or training signal.

A potential confound in the comparison above is that SA preferentially substitutes at variable positions, which tend to be more mutation-tolerant; the enrichment relative to the overall dataset rate could then reflect *where* SA mutates rather than *which* residue it selects. To separate these effects, we recomputed the enrichment against a position-matched null. For each matched substitution, the null tolerance probability is the fraction of all tested mutant amino acids at that same DMS position that are classified as tolerated; summing these per-substitution probabilities gives the number of tolerated substitutions expected if SA had chosen a random

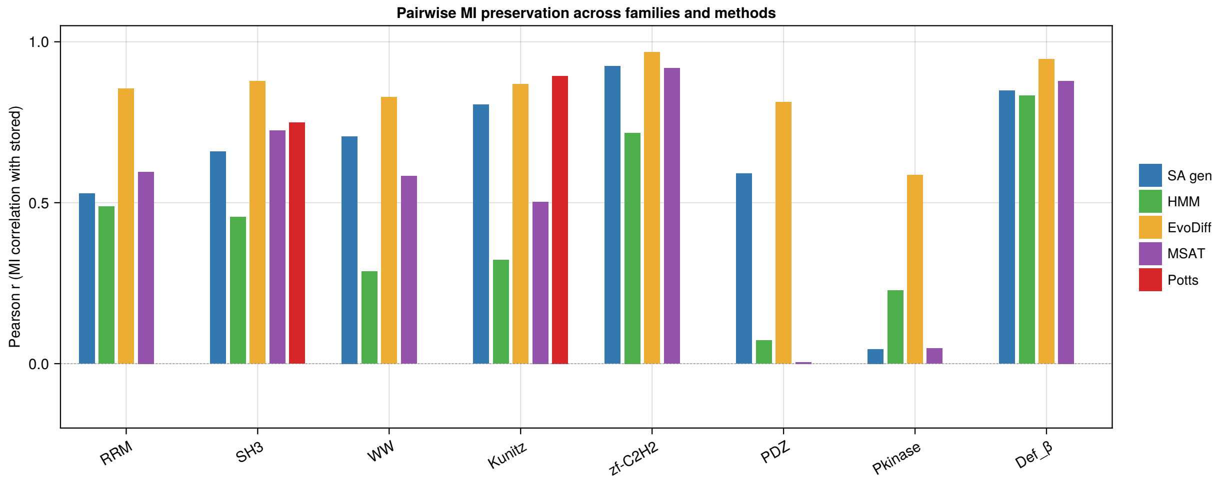

Figure S7: Pairwise MI preservation across all eight Pfam families. Each bar shows the Pearson correlation between the upper-triangle MI vector of the stored seed alignment and the MI vector of sequences generated by the indicated method. The general ordering HMM $<$ SA gen $<$ Potts $\leq$ EvoDiff reflects the information hierarchy of each model in several families, though Defensin_beta is an exception where the Potts model ($r{=}0.23$) falls below both SA and HMM. Pkinase is an outlier for SA, HMM, and MSAT owing to its high data-to-position ratio ($K{=}37$, $L{=}262$), while Potts and EvoDiff achieve high MI correlations on this family.

tested residue at each position it mutated, holding the position distribution fixed. The position-matched null rate is indeed higher than the overall rate in both families (0.843 for PDZ and 0.749 for SH3, versus 0.775 and 0.675 overall), confirming that SA does target tolerant positions. Even against this stricter null, however, SA-generated substitutions remained significantly enriched for tolerated mutations: 91.7% versus 84.3% for PDZ ($1.09\times$; Poisson-binomial $z{=}18.6$, $p < 10^{-77}$; $p < 5 \times 10^{-5}$ by a $20{,}000$-draw permutation test) and 90.4% versus 74.9% for SH3 ($1.21\times$; $z{=}23.4$, $p < 10^{-121}$). A coarser conservation-matched null, which draws comparison substitutions from positions in the same alignment-conservation quintile, gave essentially identical null rates (0.841 and 0.744) and the same conclusion (Table S9). The residue choices made by SA are therefore enriched for experimentally tolerated mutations beyond what its preference for tolerant positions alone would produce, although the enrichment is more modest than the comparison against the overall dataset rate suggests.

S12 Potts Model (plmDCA) Baseline

To assess whether pairwise coevolutionary models provide a stronger baseline than profile HMMs, we fitted a Potts model via pseudo-likelihood maximization (plmDCA) [6] to all eight Pfam seed alignments and generated 150 sequences per family by Gibbs sampling (500 sweeps per chain, initialized from random stored sequences). The Potts model captures pairwise couplings $J_{ij}(a, b)$ between all position pairs in addition to single-site fields $h_i(a)$, giving it substantially more parameters than a profile HMM: the number of coupling parameters scales as $L(L{-}1)q^2/2$, yielding data-to-parameter ratios ranging from 2.5×10^{-6} (Pkinase) to 2.0×10^{-3} (WW). Despite this high parameter count, L2 regularization ($\lambda_J{=}\lambda_h{=}0.01$) yielded well-behaved generation in most families (Table S10).

Across the eight families, the Potts model achieved comparable or better amino acid composition fidelity than SA in six of eight families, consistent with its explicit fitting of positional statistics. However, SA produced higher novelty in five of eight families, indicating that the Hopfield energy landscape supports broader exploration of sequence space than Gibbs sampling from a fitted Potts distribution. Both methods occupy an intermediate regime between profile HMMs (which produced lower family identity) and bootstrap resampling (which has zero novelty). Two families highlighted limitations for the Potts model: in Pkinase ($K{=}37$, $L{=}262$), the high data-to-parameter ratio led to low novelty (0.22 vs. 0.48 for SA) despite good compositional fidelity, suggesting the Gibbs sampler remained near stored patterns; in Defensin_beta ($K{=}45$, $L{=}36$), the Potts model produced the

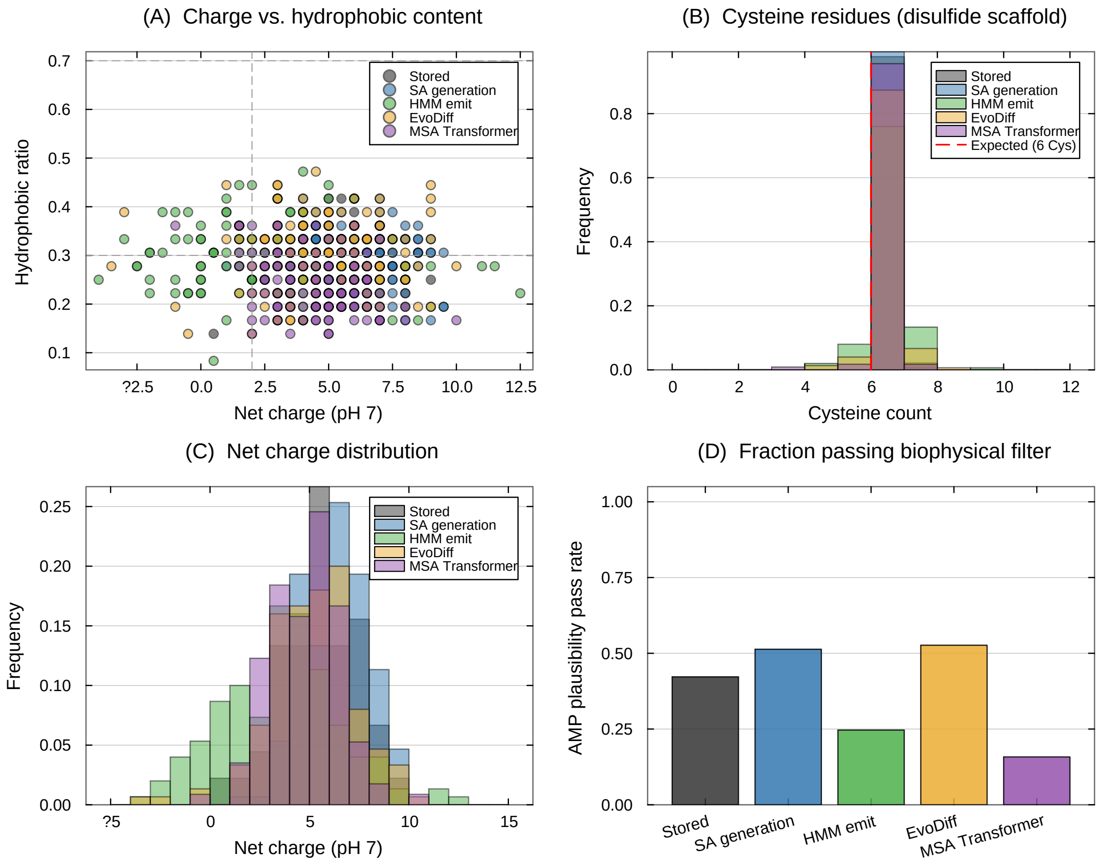

Figure S8: Biophysical properties of beta defensin sequences (PF00711). **(A)** Net charge at pH 7 vs. hydrophobic residue fraction. Sequences generated by stochastic attention (blue) overlap the stored natural sequences (gray), while HMM emit (green) and MSA Transformer (purple) drift to lower charge. Dashed lines indicate the antimicrobial plausibility thresholds (charge $\geq +2$, hydrophobic fraction 0.3–0.7). **(B)** Cysteine count distribution. Stochastic attention generation preserves the six-cysteine disulfide scaffold (red dashed line) with the same fidelity as the stored alignment. **(C)** Net charge distribution. Stochastic attention generation produces a charge profile closely matching stored sequences. **(D)** Fraction of sequences passing a minimal biophysical plausibility filter (charge $\geq +2$, hydrophobic fraction $\in [0.3, 0.7]$, ≥ 4 cysteines).

highest KL divergence of any method (KL=0.112), possibly reflecting sensitivity to the cysteine-rich, low-entropy character of this family. The main distinction between the two approaches is that the Potts model required fitting hundreds of thousands to millions of parameters via L-BFGS-B optimization, followed by Gibbs sampling (total wall time: 0.5–52 minutes per family), while SA required no parameter fitting and generated sequences in seconds. The broadly similar output quality across the remaining six families supports the conclusion that some of the correlational structure captured by pairwise couplings is also present in the Hopfield energy landscape, which implicitly encodes higher-order correlations through the softmax attention operation.

S13 Sampling Diagnostics

To characterize the mixing properties of the Langevin sampler and justify the burn-in and thinning parameters used throughout the paper, we computed autocorrelation functions and effective sample sizes (ESS) from the Hopfield energy trace $E(\boldsymbol{\xi}_t)$ along each chain. For each family, we ran 10 independent unadjusted Langevin algorithm (ULA) and Metropolis-adjusted Langevin algorithm (MALA) chains of $T{=}5{,}000$ steps at the generation temperature $\beta_{\text{gen}} \approx 2\beta^*$, recording the full trajectory. We computed the normalized autocorrelation function of the post-burn-in energy trace (iterations $2{,}001$–$5{,}000$), estimated the integrated autocorrelation time τ_{int} using the initial positive sequence estimator with a cutoff at $\rho(\tau) < 0.05$, and derived the effective sample size as $\text{ESS} = n/\tau_{\text{int}}$ where $n{=}3{,}000$ is the number of post-burn-in samples.

Table S11 reports the results across all eight families. The integrated autocorrelation time ranged from 76 to 122 iterations for ULA and from 82 to 131 for MALA, yielding 32–42 effective samples per chain (ULA) and 27–39 per chain (MALA). The thinning interval of 100 is thus well matched to the autocorrelation time ($\text{thin}/\tau_{\text{int}} = 0.8$–$1.3$), yielding retained samples that are approximately independent. Across 30 chains, the total effective sample size is approximately $30 \times 35 \approx 1{,}050$ independent draws from the Boltzmann distribution, from which we retain 150 decoded sequences (5 per chain). The burn-in of $2{,}000$ iterations provides a 3.2–$6.6\times$ margin over the empirical convergence point (304–617 iterations), supporting the use of this burn-in before sampling begins. The MALA acceptance rates (99.6–99.8%) show that the Metropolis correction rejects fewer than 0.4% of proposed moves at step size $\alpha{=}0.01$, so the unadjusted and adjusted updates rarely differ in this regime. A high acceptance rate bounds rather than directly quantifies the stationary-distribution bias of ULA; we therefore report it only as evidence that the two samplers take nearly identical steps at this step size, and adopt ULA for its computational efficiency.

S14 Initialization Sensitivity

The multi-chain protocol described in Materials and Methods initializes each Langevin chain near a randomly selected stored pattern ($\boldsymbol{\xi}_0 = \mathbf{m}_k + \sigma_{\text{init}}\boldsymbol{\epsilon}$, $\sigma_{\text{init}} = 0.01$). To test whether this choice affects the generated ensemble, we repeated the full SA generation pipeline with random initialization on the unit sphere: each chain was started from $\boldsymbol{\xi}_0 = \mathbf{z}/\|\mathbf{z}\|$ with $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, placing the initial point at a random location unrelated to any stored pattern. All other parameters were held fixed.

The results were consistent across initialization strategies: random sphere initialization produced output indistinguishable from stored-pattern initialization across all eight families (Table S12). The maximum absolute difference in KL divergence was 0.0003 (RRM), and all other families showed differences below 10^{-4} in every metric. This insensitivity is expected from the theory: at $\beta_{\text{gen}} \approx 2\beta^*$, the energy landscape has well-defined basins near stored patterns, and the dissipative drift term $-\alpha\nabla E(\boldsymbol{\xi})$ rapidly pulls any initial condition toward the nearest basin. The 2,000-step burn-in (at step size $\alpha{=}0.01$, corresponding to 20 effective time units) is long relative to the observed convergence point. The result indicates that stored-pattern initialization is a computational convenience (reducing burn-in time) rather than a requirement, and that the generated ensemble is not an artifact of the initialization strategy.

S15 Near-Duplicate Analysis

A key concern for any generative model operating on small datasets is whether the generated sequences are trivially close to stored patterns, effectively copying inputs with minor perturbations rather than producing novel outputs. To quantify the degree of near-duplication, we computed the full pairwise sequence identity between every SA-generated sequence ($S{=}150$) and every stored sequence (K) for each family, yielding a $150{\times}K$ identity matrix. For each generated sequence, we recorded the maximum identity to any stored sequence (nearest neighbor) and the minimum number of amino acid substitutions.

Table S13 reports the near-duplicate statistics across all eight families. Zero generated sequences across all eight families have $> 95\%$ or $> 90\%$ identity to any stored sequence, and zero sequences are within 2 substitutions of a stored pattern. Only in the Defensin_beta family ($L{=}36$, the shortest alignment) do 6% of generated sequences exceed 80% identity to a stored sequence, which corresponds to at most 7 substitutions in a 36-residue sequence. The mean nearest-neighbor identity for generated sequences (0.515–0.655) is substantially higher than the mean within-family pairwise identity (0.217–0.368), indicating that generated sequences are closer to individual stored patterns than stored patterns are to each other on average, but not trivially close. These results support the interpretation that SA generates sequences in intermediate regions of sequence space rather than copying stored patterns.

The nearest sequence identity reported for generated sequences ("SeqID") is a nearest-neighbor quantity: the identity of each generated sequence to its closest stored pattern. The appropriate natural reference is therefore the within-family nearest-neighbor identity, computed by leave-one-out as each stored sequence's identity to its closest neighbor in the same family, not the mean pairwise identity ("Stored ID" above), which averages over all distant pairs and is consequently much lower. Table S14 reports this nearest-neighbor envelope. Mean nearest-neighbor identity among natural sequences ranges from 0.38 (Pkinase) to 0.60 (WW), with 10th–90th percentile envelopes spanning roughly 0.29–0.74, far above the 0.22–0.37 mean pairwise identity. Generated SeqID (0.51–0.66) falls within each family's full nearest-neighbor range for seven of eight families; in Pkinase, generated SeqID (0.515) slightly exceeds the natural maximum (0.454), consistent with this family's strongly consensus-proximal generation (Section S6). By contrast, profile HMM emit and the MSA Transformer reach nearest-neighbor identities of 0.09–0.21 in several families, below the minimum nearest-neighbor identity of any natural sequence in those families (e.g., 0.22–0.28 for Pkinase, PDZ, and RRM), supporting the conclusion that their high novelty reflects drift outside the family rather than family-consistent exploration. All 150 generated sequences are distinct in every family, with mean pairwise sequence identity among generated sequences of 0.35–0.65 (mean pairwise Hamming distance 0.36–0.65), a sequence-space diversity measure that does not rely on high-dimensional cosine distance.

S16 Wall-Clock Timing Comparison

Table S15 reports wall-clock generation times for 150 sequences across all eight families. Stochastic attention and profile HMM emit were timed on a single CPU core (Apple M2, 16 GB RAM) with all preprocessing included (PCA construction, phase transition detection, sampling, and decoding for SA; HMM building and sequence emission for HMM emit). EvoDiff and MSA Transformer times were measured on the same hardware (CPU only, no graphics processing unit), and Potts model (plmDCA) times include both parameter fitting and Gibbs sampling.

Stochastic attention generated 150 sequences in 0.2–4.6 s across all eight families, with the variation driven primarily by the number of stored patterns K (which determines the cost of the softmax attention at each Langevin step). The largest family (WW, $K{=}420$) required 4.6 s, while all families with $K < 100$ completed in under 1 s. Profile HMM emit was consistently the fastest method (0.02–0.09 s), as expected for a method that samples from a precomputed position-specific model with no iterative computation. EvoDiff required 2–14 hours on CPU depending on sequence length (GPU inference would reduce these times substantially, though EvoDiff's intended use case assumes GPU availability), and the MSA Transformer required 1–8 hours for 50 rounds of iterative masked language model sampling. In these CPU-only runs, SA was 3–5 orders of magnitude faster than EvoDiff and the MSA Transformer, and 2–3 orders of magnitude faster than Potts model fitting, while requiring no GPU and no pretraining.

S17 Consensus-with-Noise Baseline

A natural concern is that SA may function as a "consensus generator" that merely perturbs the family consensus sequence. To test this directly, we implemented a control that adds calibrated noise to the consensus sequence in PCA space: for each family, we computed the position-wise consensus sequence (most frequent amino acid per column), one-hot encoded it, projected it into the same PCA space used by SA, and generated 150 sequences by adding isotropic Gaussian noise with standard deviation matched to the empirical spread of stored patterns in PCA space. We then decoded each noisy PCA vector via inverse PCA and argmax, and evaluated the resulting sequences using the same metrics as SA (Table S16).

The comparison reveals a characteristic signature that distinguishes SA from consensus perturbation. The consensus-with-noise baseline produces sequences with systematically higher novelty (0.63–0.79 vs. 0.40–0.65 for SA) and lower sequence identity to stored patterns (0.44–0.62 vs. 0.52–0.66), reflecting the fact that isotropic noise in PCA space scatters sequences away from all stored patterns rather than interpolating between them. The KL divergence pattern is mixed: the consensus baseline achieves lower KL in some families (e.g., RRM: 0.013 vs. 0.060) but higher KL in others (e.g., zf-C2H2: 0.095 vs. 0.013), showing that isotropic noise can either improve or degrade composition depending on whether the family's covariance aligns with the noise directions. The key distinction is that SA generation maintains higher sequence identity to stored patterns while achieving substantial novelty, indicating that it explores the family's sequence space in a structured way that respects the correlational landscape, whereas consensus perturbation explores isotropically without regard to the family's covariance structure.

S18 Column-Permuted Alignment Control

To test directly whether SA captures information beyond marginal per-position amino acid frequencies, we constructed a control alignment for each family by independently shuffling each column. This procedure preserves the exact per-position amino acid frequency distribution while destroying all inter-position correlations (covariation, structural contacts, epistatic couplings). We then ran the full SA pipeline on the permuted alignment and compared the output against SA run on the real alignment (Table S17).

The results show a consistent pattern across all eight families. Stochastic attention on the real alignment produces sequences with lower novelty but higher sequence identity to stored patterns compared to stochastic attention on the permuted alignment. This is expected: the real alignment contains inter-position correlations that constrain the energy landscape and keep generated sequences closer to family-typical regions, whereas the permuted alignment provides only position-independent frequency information, yielding an energy landscape that allows more isotropic exploration. The MI correlation differences between real and permuted SA are small but consistently favor the real alignment in seven of eight families (the exception being zf-C2H2, where the difference is < 0.01), with the magnitude of the difference ranging from 0.01 to 0.08. The larger differences in novelty and sequence identity indicate that the inter-position correlations present in real alignments shape the generated ensemble in ways that marginal frequencies alone cannot reproduce.

S19 PCA Variance Threshold Sensitivity

To assess whether the default 95% variance threshold for PCA dimensionality reduction is a fragile design choice, and to identify the minimum information content needed for reliable generation, we repeated the full SA generation pipeline at eleven retained-variance thresholds spanning 50% to 99%. For each family and threshold, we constructed the memory matrix, identified the critical temperature β^* via the entropy inflection criterion, and generated 150 sequences (30 chains $\times$ 5 samples) at $\beta_{\text{gen}} = \max(\lceil 2\beta^* \rceil, 5)$. Table S18 reports six representative thresholds for all eight families.

The results reveal two regimes separated by a transition near 70–75% retained variance. Above this threshold, generation quality is robust: KL divergence remained below 0.09 in every condition, novelty broadly increased with the variance threshold (higher d provides more dimensions for exploration), and nearest sequence identity was essentially constant within each family. The critical temperature β^* was largely insensitive to the threshold: in six of eight families, β^* was identical across all eleven thresholds, reflecting the discrete resolution of the entropy inflection search grid. Below 70% retained variance, compositional fidelity began to degrade in the more diverse families. The clearest example was the WW domain ($K{=}420$), where KL divergence was 0.03 at 80% but rose to 0.15 at 50%, a five-fold increase indicating that discarded principal components encoded position-specific conservation patterns needed for faithful generation. By contrast, families with lower intrinsic diversity (PDZ, Pkinase, Defensin_beta) were nearly flat across the entire range because their first few principal components already captured the relevant variation. The transition is thus family-dependent, governed by the alignment's spectral structure: families whose eigenvalue spectrum decays slowly require more principal components to preserve compositional fidelity, while compact families tolerate aggressive compression. These findings establish that the 95% default lies well within the robust regime and that practitioners can safely vary the threshold between 75% and 99% without substantially affecting output quality.

S20 Sequence Encoding, PCA Projection, and Decoding

Stochastic attention operates on continuous vectors on the unit sphere $\mathbb{S}^{d-1}$, while protein sequences are discrete objects over a 20-letter amino acid alphabet. Bridging the two requires three transformations (one-hot encoding, PCA dimensionality reduction, and unit-norm projection), and the reverse path (inverse PCA followed by argmax decoding) recovers discrete sequences from continuous samples. Given a cleaned alignment of K

sequences, each of length L amino acids over the standard 20-letter alphabet $\mathcal{A} = \{\mathsf{A}, \mathsf{R}, \mathsf{N}, \ldots, \mathsf{V}\}$, we encode each sequence as a binary vector in $\mathbb{R}^{20L}$ by concatenating per-position indicator vectors:

$$\mathbf{x}_k = \left[\mathbf{e}_{a_1^{(k)}},\ \mathbf{e}_{a_2^{(k)}},\ \ldots,\ \mathbf{e}_{a_L^{(k)}}\right]^\top \in \{0,1\}^{20L}, \tag{S1}$$

where $\mathbf{e}_a \in \{0,1\}^{20}$ is the standard basis vector for amino acid a. The full-dimensional representation is $d_{\text{full}} = 20L$, which ranges from 460 (zf-C2H2, $L{=}23$) to $5{,}240$ (Pkinase, $L{=}262$) across the eight families studied. This encoding is lossless and treats each position-amino acid pair as an independent coordinate, making no assumptions about residue similarity or physicochemical properties. Gap and non-standard-residue positions are encoded as the all-zero 20-vector, so they contribute nothing to the inner products that drive retrieval. On decoding, a position is assigned a gap only if its reconstructed 20-channel block is numerically zero (maximum entry below 10^{-10}); because inverse PCA produces dense reconstructions, every decoded position resolved to a standard amino acid, and decoded sequences therefore contained no gaps, consistent with the 100% valid-amino-acid rate. Sequence identity is computed over non-gap positions only.

The full one-hot space is high-dimensional relative to the number of stored patterns; the ratio d_{full}/K ranges from 1.5 (WW) to 142 (Pkinase). This creates two problems for SA. First, in the modern Hopfield energy, the similarity scores $e_k = \mathbf{m}_k^\top \boldsymbol{\xi}$ have variance $1/d$ for a random query on $\mathbb{S}^{d-1}$; when d is large, scores concentrate tightly around zero and high β is required to differentiate among stored patterns, which makes the energy landscape nearly flat and renders Langevin sampling inefficient. Second, the one-hot encoding introduces substantial redundancy: at each position, only 1 of 20 coordinates is nonzero. Principal component analysis removes this redundancy by projecting onto the directions of actual variation. We compute the economy singular value decomposition of the centered data matrix $\tilde{\mathbf{X}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ and select the smallest p such that the retained variance fraction $\rho(p) = \sum_{j=1}^{p} \sigma_j^2 / \sum_j \sigma_j^2 \geq 0.95$. Among all rank-$p$ linear projections, PCA minimizes the mean squared reconstruction error, making it the optimal linear dimensionality reduction for preserving the structure of the data. Across the eight families, the retained PCA dimension d ranged from 34 (Pkinase) to 186 (WW).

After PCA, we normalize each projected vector to unit L_2 norm: $\mathbf{m}_k = \mathbf{z}_k/\|\mathbf{z}_k\|_2 \in \mathbb{S}^{d-1}$, which ensures that the similarity scores lie in $[-1, 1]$ and that the concentration-of-measure results hold exactly for random unit-sphere probes. To decode a Langevin sample $\boldsymbol{\xi} \in \mathbb{R}^d$ back to an amino acid sequence, we apply inverse PCA ($\hat{\mathbf{x}} = \bar{\mathbf{x}} + \mathbf{W}_d\boldsymbol{\xi}$) followed by per-position argmax decoding: for each position ℓ, the amino acid with the largest coordinate in the 20-dimensional sub-vector is selected. This deterministic decoding produced 100% valid amino acids across all families and conditions. Several properties make PCA particularly well suited for the modern Hopfield energy: linearity (both projection and inverse are matrix multiplications, keeping the score function in closed form), variance-optimality (it captures maximum family-level variation in the fewest dimensions), and parameter-free determinism (the projection is uniquely defined by the SVD and variance threshold, with no hyperparameters to tune). Algorithm 1 summarizes the complete pipeline.

S21 Analysis of the Critical Temperature

We developed an analytical framework to understand the empirical relationship $\beta^* \approx 1.52 + 0.28\sqrt{d}$ reported in Materials and Methods, starting from exact results on similarity variance, building a Gaussian mean-field prediction, and then diagnosing why the mean-field coefficient overshoots and how stored-pattern self-similarity resolves the discrepancy. After one-hot encoding and PCA projection to d dimensions, the stored patterns $\mathbf{m}_k$ are normalized to lie on $\mathbb{S}^{d-1}$. For a random query $\boldsymbol{\xi}$ drawn uniformly from $\mathbb{S}^{d-1}$, each similarity $e_k = \mathbf{m}_k^\top \boldsymbol{\xi}$ has mean zero and variance $1/d$, regardless of the structure of $\mathbf{m}_k$, a consequence of the rotational invariance of the uniform distribution on the sphere. We verified this numerically for all eight families using 1,000 random probes per family: the ratio $\sigma^2 \cdot d$ is 1.000 ± 0.003 across all families and WW subsamples (Fig. S9B,F), and the excess kurtosis is mildly negative (-0.06 to -0.33), consistent with sub-Gaussian behavior on $\mathbb{S}^{d-1}$. The characteristic similarity scale is therefore $\sigma = 1/\sqrt{d}$, and since β enters the softmax as βe_k, the entropy inflection must occur when $\beta\sigma = \mathcal{O}(1)$, giving $\beta^* \sim \sqrt{d}$.

These observations motivated a Gaussian mean-field model: if the K similarity scores were i.i.d. draws from $\mathcal{N}(0, 1/d)$, the softmax entropy $H(\tau) = -\sum_k p_k \log p_k$ with $p_k \propto \exp(\tau z_k)$ undergoes an entropy inflection between the uniform-attention regime ($H = \log K$, $\tau = 0$) and the pattern-selective regime ($H \to 0$, $\tau \to \infty$). We defined the dimensionless entropy inflection τ^* using the same entropy-inflection criterion as

Algorithm 1 Stochastic Attention Protein Sequence Generation

Require: Seed alignment $\mathbf{S} \in \mathcal{A}^{K\times L}$, variance threshold $\rho_{\min} = 0.95$, step size $\alpha = 0.01$, chains $N_c = 30$, iterations $T = 5000$, burn-in $T_b = 2000$, thinning $\Delta t = 100$, samples/chain $s = 5$

Ensure: Generated sequences $\{\mathbf{g}_1, \ldots, \mathbf{g}_S\} \subset \mathcal{A}^L$, $\;S = N_c \cdot s$

Encoding:

One-hot encode: $\mathbf{X} \leftarrow [\mathrm{onehot}(\mathbf{S}_{1,:}), \ldots, \mathrm{onehot}(\mathbf{S}_{K,:})] \in \{0,1\}^{20L\times K}$

Center: $\bar{\mathbf{x}} \leftarrow \frac{1}{K}\sum_k \mathbf{X}_{:,k}$; $\;\tilde{\mathbf{X}} \leftarrow \mathbf{X} - \bar{\mathbf{x}}\mathbf{1}_K^\top$

SVD: $\tilde{\mathbf{X}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$

Select d: smallest p such that $\sum_{j=1}^{p}\sigma_j^2/\sum_j \sigma_j^2 \geq \rho_{\min}$; set $\mathbf{W}_d \leftarrow \mathbf{U}_{:,1:d}$

Project and normalize: $\mathbf{m}_k \leftarrow \mathbf{W}_d^\top(\mathbf{X}_{:,k} - \bar{\mathbf{x}})/\|\mathbf{W}_d^\top(\mathbf{X}_{:,k} - \bar{\mathbf{x}})\|$ for $k = 1, \ldots, K$

Assemble memory: $\hat{\mathbf{X}} \leftarrow [\mathbf{m}_1, \ldots, \mathbf{m}_K] \in \mathbb{R}^{d\times K}$

Temperature selection:

Set $K_p \leftarrow \min(K, 20)$ and evaluate 50 log-spaced β values in $[0.1, 500]$

For $i = 1, \ldots, K_p$, compute $\mathbf{p}_i(\beta) = \mathrm{softmax}(\beta\,\hat{\mathbf{X}}^\top\mathbf{m}_i)$ and $H_i(\beta) = -\sum_{k=1}^{K} p_{ik}(\beta)\log p_{ik}(\beta)$

Set β^* to the entropy inflection point of $\bar{H}(\beta) = K_p^{-1}\sum_i H_i(\beta)$ on this grid

Set $\beta_{\mathrm{gen}} \leftarrow \max\big(\lceil 2\beta^*\rceil,\ 5\big)$

Multi-chain Langevin sampling:

$\mathcal{G} \leftarrow \emptyset$

for $c = 1, \ldots, N_c$ **do**

  Draw $k_c \sim \mathrm{Uniform}\{1, \ldots, K\}$

  $\boldsymbol{\xi}_0 \leftarrow \mathbf{m}_{k_c} + 0.01\,\boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$

  for $t = 0, \ldots, T-1$ **do**

    $\mathbf{a}_t \leftarrow \mathrm{softmax}(\beta_{\mathrm{gen}}\,\hat{\mathbf{X}}^\top\boldsymbol{\xi}_t)$

    $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$

    $\boldsymbol{\xi}_{t+1} \leftarrow (1-\alpha)\,\boldsymbol{\xi}_t + \alpha\,\hat{\mathbf{X}}\,\mathbf{a}_t + \sqrt{2\alpha/\beta_{\mathrm{gen}}}\,\boldsymbol{\epsilon}_t$

  end for

  Collect s evenly spaced samples from $\{\boldsymbol{\xi}_{T_b+1}, \boldsymbol{\xi}_{T_b+\Delta t+1}, \ldots, \boldsymbol{\xi}_T\}$ into $\mathcal{G}$

end for

Decoding:

for each $\boldsymbol{\xi} \in \mathcal{G}$ **do**

  Inverse PCA: $\hat{\mathbf{x}} \leftarrow \bar{\mathbf{x}} + \mathbf{W}_d\,\boldsymbol{\xi} \in \mathbb{R}^{20L}$

  Argmax decode: $g_\ell \leftarrow \arg\max_{a\in\mathcal{A}}\ \hat{x}_{20(\ell-1)+a}$ for $\ell = 1, \ldots, L$

end for

return $\{\mathbf{g}_1, \ldots, \mathbf{g}_S\}$

the empirical β^* search and computed it numerically by averaging over 500 Gaussian realizations for each $K \in \{10, 20, \ldots, 1000\}$. We found that τ^* is nearly constant at ≈ 1.6 for $K \geq 30$, with only weak logarithmic growth (Fig. S9A). Translating back to physical units, the Gaussian prediction $\beta^* = \tau^*\sqrt{d} \approx 1.6\sqrt{d}$ has the correct $\sqrt{d}$ scaling and achieves Pearson $r = 0.98$ with the empirical values, but systematically overpredicts β^* by a factor of ≈ 4.

This overprediction arises because the empirical β^* is computed from stored-pattern probes $\mathbf{m}_i$, not from random queries. When $\boldsymbol{\xi} = \mathbf{m}_i$, the self-similarity score $\mathbf{m}_i^\top\mathbf{m}_i = 1$ is an outlier relative to the cross-similarities $\mathbf{m}_j^\top\mathbf{m}_i$ for $j \neq i$, which have magnitudes $\mathcal{O}(1/\sqrt{d})$. The resulting gap $\Delta = 1 - \max_{j\neq i}\mathbf{m}_j^\top\mathbf{m}_i$ ranges from 0.55 to 0.82 across families (Fig. S11E), meaning the softmax concentrates on the self-pattern at much lower β than would be needed for a random query. Substituting the cross-similarity standard deviation σ_{cross} for $1/\sqrt{d}$ fails decisively ($R^2 = -9.0$; Fig. S11D), because the cross-similarity distribution is heavy-tailed (excess kurtosis 6–25) and the self-similarity outlier violates the Gaussian assumption. The fitted model $\beta^* = 1.52 + 0.28\sqrt{d}$ nonetheless remains robust: the $\sqrt{d}$ scaling is set by concentration of measure, and the reduced coefficient 0.28 (vs. $\tau^* \approx 1.6$) encapsulates the combined effect of self-similarity anchoring and the protein family correlation structure.

To quantify uncertainty in the fitted regression, we performed a nonparametric bootstrap analysis: resampling the 33 observations (8 families plus 25 WW scaling replicates) with replacement 10,000 times yielded bootstrap

SE of 0.07 for the intercept and 0.007 for the slope, with 95% confidence intervals of $[1.39, 1.68]$ and $[0.266, 0.294]$, respectively. The bootstrap R^2 distribution has a median of 0.969 (95% CI $[0.940, 0.987]$). Because 25 of the 33 data points come from WW subsamples, we also refit the model using only the eight Pfam families: the resulting fit, $\beta^* = 1.64 + 0.278\sqrt{d}$ (R^2=0.95), is nearly identical, indicating that the result is not driven by overrepresentation of a single family. Leave-one-family-out cross-validation achieved R^2=0.96 (root mean square error, RMSE=0.18) on the full dataset and R^2=0.93 (RMSE=0.19) on the 8-family subset, with no individual family acting as an outlier (Table S19). Two cautions attend the pooled fit. First, 25 of the 33 observations are nested WW subsamples and are therefore not independent; the eight-family-only fit above is the more conservative estimate, and the leave-one-family-out results confirm out-of-sample generalization. Second, the empirical β^* is identified on a fixed grid of 50 logarithmically spaced values over $[0.1, 500]$ (multiplicative spacing ≈ 1.19); the eight family values land on consecutive grid points ($\beta^* \in \{3.23, 3.85, 4.58, 5.45\}$), so β^* is resolved only to within $\approx$19%, and the reported R^2 is computed against these grid-resolved targets. A finer grid would refine the point estimates but is not expected to alter the $\sqrt{d}$ scaling, which follows from concentration of measure rather than from the grid.

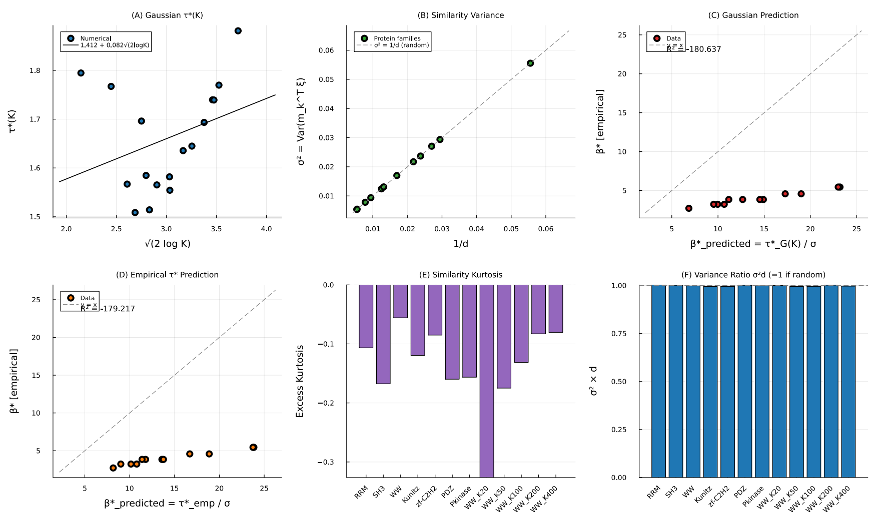

Figure S9: **Analytical investigation of β^*.** (A) Dimensionless softmax entropy inflection $\tau^*(K)$ for Gaussian scores. (B) Similarity variance σ^2 vs. $1/d$ for random probes; all points lie on the identity line. (C) Gaussian prediction $\beta^*_{\text{pred}} = \tau^*_G(K)/\sigma$ vs. empirical β^*; correct trend ($r = 0.98$) but overshoots by $\sim 4\times$. (D) Same as (C) using the empirical τ^*; overshoot persists. (E) Excess kurtosis of the similarity distribution. (F) The ratio $\sigma^2 d$ across all datasets, showing $\sigma^2 d \approx 1$.

S22 Regularity and Convergence Properties

The modern Hopfield energy has Lipschitz-continuous gradient with constant at most $L = 1 + \beta\sigma^2_{\max}/2$, where $\sigma_{\max} = \|\hat{\mathbf{X}}\|_{\text{op}}$ for the normalized memory matrix, and satisfies the dissipativity condition $\langle \nabla E(\boldsymbol{\xi}), \boldsymbol{\xi} \rangle \geq \|\boldsymbol{\xi}\|^2/2 - M^2/2$ with $M = \max_k \|\mathbf{m}_k\|$. These regularity properties determine the convergence regime of the Langevin sampler. When $\beta\sigma^2_{\max} < 2$, this sufficient bound implies convexity of the energy and ULA iterates converge to p_β in Wasserstein-2 distance at a geometric rate [7, 8]. In the protein setting, operating temperatures satisfy $\beta\sigma^2_{\max} \gg 2$, placing the sampler outside this globally convex regime, where the energy landscape can have multiple metastable basins corresponding to stored patterns or clusters thereof. Despite this non-convexity, the multi-chain protocol with long burn-in and thinning yields well-mixed chains, and the initialization sensitivity analysis indicates that the sampled distribution is insensitive to starting conditions.

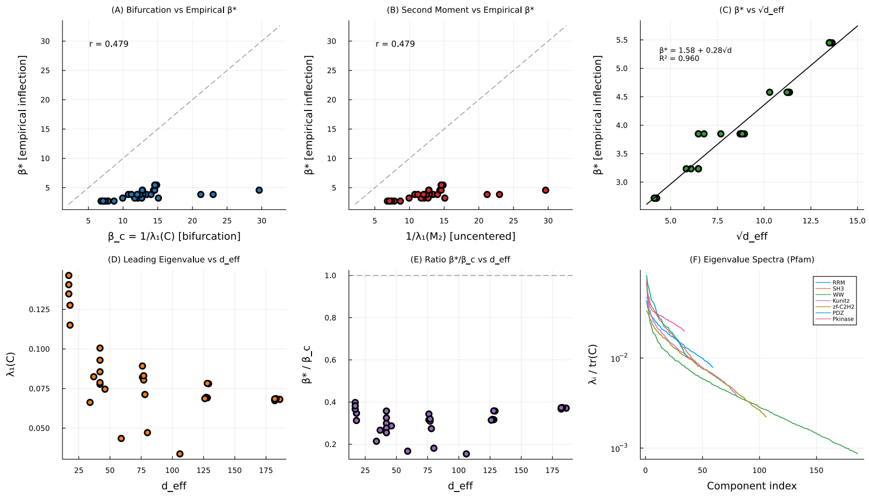

Figure S10: **Bifurcation-based prediction of β^*.** (A) $\beta_c = 1/\lambda_1(\mathbf{C})$ vs. empirical β^* ($r{=}0.479$); bifurcation overshoots by 3–$5\times$. (B) Second-moment predictor performance is identical. (C) β^* vs. $\sqrt{d}$ with fitted model ($R^2{=}0.960$). (D) Leading eigenvalue vs. d; no consistent relationship. (E) Ratio β^*/β_c showing that the bifurcation estimate systematically overestimates. (F) Normalized eigenvalue spectra for all eight families.

S23 Pfam Family Size Census

To verify the claims about Pfam family sizes made in the main text, we performed a census of all Pfam-A families using release 36.0, the final standalone Pfam release before its integration into InterPro [9]. We downloaded both the seed alignment file (`Pfam-A.seed.gz`) and the full alignment file (`Pfam-A.full.gz`) from the European Bioinformatics Institute FTP archive and parsed each Stockholm-format entry to count the number of unique sequences per family. Seed alignments are small, manually curated sets of representative sequences that Pfam curators selected to capture the diversity of each family while maintaining alignment quality. These seed alignments serve as the input to HMMER, which builds a profile HMM and searches UniProtKB to produce the full alignment. The full alignment therefore contains every protein sequence that HMMER can confidently assign to the family, making it substantially larger but also noisier: it includes automatically identified homologs, sequence fragments, and partial matches that were not subject to manual curation. Because our method and most alignment-based generative approaches operate on curated seed alignments rather than raw database hits, the seed alignment sizes are the operationally relevant quantities.

Table S20 summarizes the resulting distributions across all 20,795 Pfam-A families. The median seed alignment contains only 22 sequences (interquartile range: 7–65), and 69.2% of families have fewer than 50 seed sequences. Nearly half (47.5%) have fewer than 20. The full alignments are considerably larger, with a median of 841 sequences (interquartile range: 127–2,973), reflecting the breadth of automated homology detection across UniProtKB. Nevertheless, even at the full-alignment level, 22.7% of families have fewer than 100 sequences, and the distribution remains heavily right-skewed: a small number of ubiquitous superfamilies (protein kinases, immunoglobulins, zinc fingers) contain hundreds of thousands of members, while the long tail of specialized families remains data-poor. Figure S12 shows the empirical cumulative distribution functions for both seed and full alignment sizes on a logarithmic scale.

These distributions confirm that the majority of protein families fall well below the data requirements of current deep generative architectures, whether measured by curated seed alignments or by the broader full-alignment census, motivating training-free approaches such as stochastic attention.

S24 Software Versions, Random Seeds, and Reproducibility

All experiments were run on an Apple M2 workstation (16 GB RAM). The stochastic-attention sampler, baseline evaluation, scaling study, sampling diagnostics, and all metric computations were implemented in Julia 1.12.5;

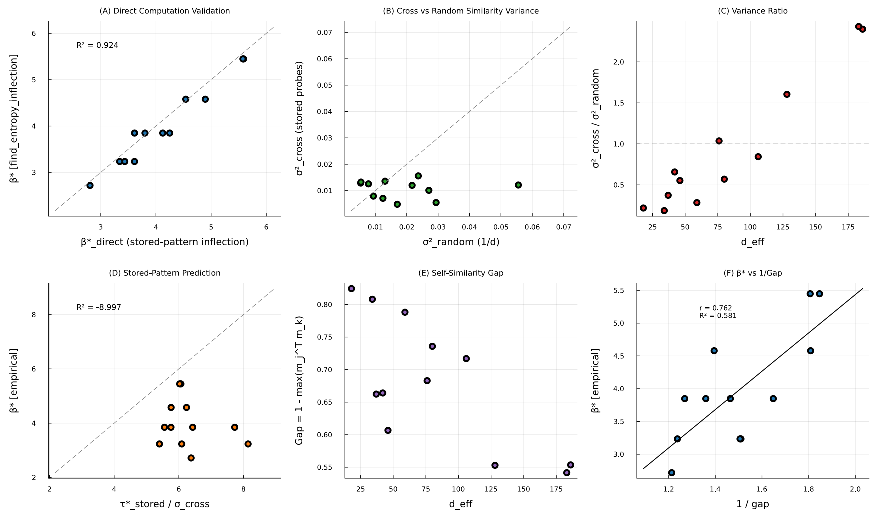

Figure S11: **Stored-pattern probe analysis.** (A) β^* from stored-pattern entropy inflection vs. the value from the entropy-inflection search. (B) Cross-similarity variance (stored probes) vs. random-probe variance. (C) Variance ratio vs. d. (D) The decomposition $\tau^*_{\text{stored}}/\sigma_{\text{cross}}$ fails as a predictor ($R^2 = -9.0$). (E) Self-similarity gap Δ vs. d. (F) β^* vs. $1/\Delta$ ($r = 0.76$).

the principal packages and their versions were Flux 0.16.9, NNlib 0.9.33, MultivariateStats 0.10.4 (principal component analysis), Distributions 0.25.123, StatsBase 0.34.10, Clustering 0.15.8, DataFrames 1.8.1, CSV 0.10.16, and Plots 1.41.6 with StatsPlots 0.15.8. The complete Julia environment is pinned by the `Project.toml` and `Manifest.toml` files in the repository. The deep-learning baselines and validators were run in Python 3.12 using the following published model checkpoints: ESM2-650M (`esm2_t33_650M_UR50D`) for pseudo-perplexity scoring; the MSA Transformer (`esm_msa1b_t12_100M_UR50S`); EvoDiff (`MSA_OA_DM_RANDSUB`); and ESMFold together with AlphaFold2 for structure prediction, the latter run through ColabFold with model type `alphafold2_ptm`, three recycles, one model, and seed 0. Profile HMMs were built and sampled with HMMER 3.4, structural superpositions used TM-align [10], and the Potts baseline used a custom pseudo-likelihood maximization (plmDCA) implementation included in the repository. The exact versions of the Python dependencies (PyTorch, `fair-esm`, EvoDiff, ColabFold, NumPy, SciPy, Biopython) and of the TM-align binary are recorded in the environment specification archived with the versioned software record described in the main-text Data, Materials, and Software Availability statement.

Random seeds were fixed for every stochastic step so that each experiment is reproducible. Table S21 lists the seeds used; per-chain seeds use a fixed base offset plus the chain index c.

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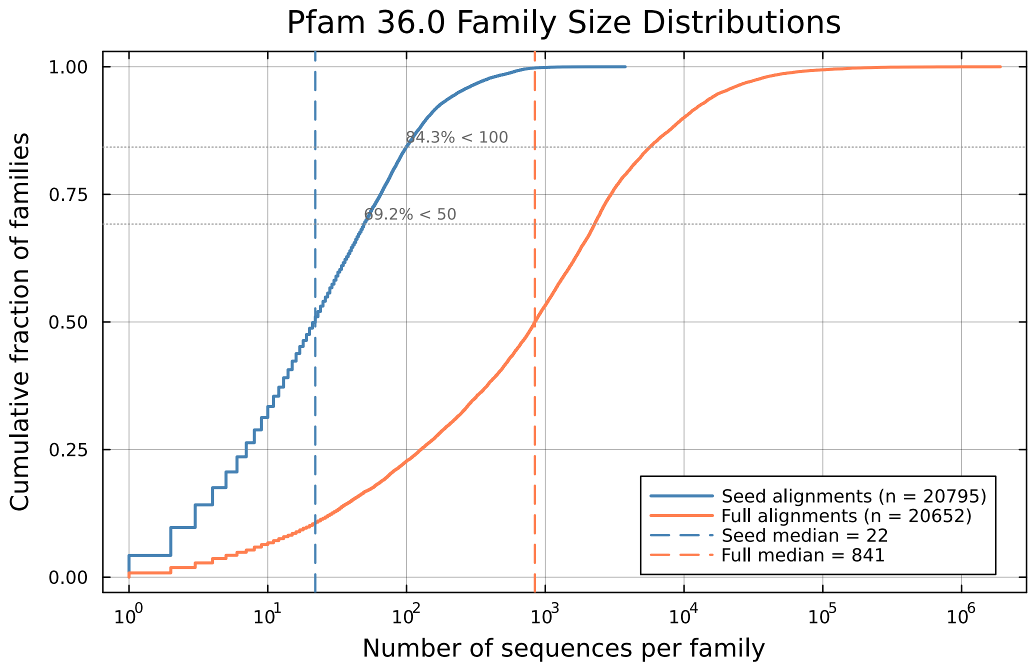

Figure S12: Empirical cumulative distribution functions of family size for Pfam 36.0 seed alignments (blue) and full alignments (red). Dashed vertical lines mark medians. The majority of seed alignments contain fewer than 50 sequences.

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Table S1: Generation metrics across eight Pfam families (mean $\pm$ SE). The label SA gen denotes the generation regime ($\beta \approx 2\beta^{*}$), and SA ret denotes the retrieval regime ($\beta \approx 20\beta^{*}$). Baseline abbreviations are BS, bootstrap; GP, Gaussian perturbation; and CC, convex combination. The KL column reports amino acid composition KL divergence ($\downarrow$ lower is better; SE via 1,000-fold bootstrap). Novelty is $1 - \max_k \cos(\hat{\boldsymbol{\xi}}, \mathbf{m}_k)$ in PCA space ($\uparrow$ higher is better). SeqID is nearest sequence identity to a stored pattern after decoding ($\downarrow$ lower is better). Bold indicates the best method per family and metric. The KL SE here is the $1{,}000$-fold bootstrap SE; because KL divergence is non-negative and near zero for several families, this bootstrap distribution is right-skewed, so the symmetric SE can exceed the point estimate (e.g., zf-C2H2, where a percentile interval would be more appropriate). The per-family comparison tables (Tables S2 and S10) instead report the SE across 30 sub-blocks of five sequences; the SA-generation KL point estimates are identical across tables, and only the SE estimator differs.

Family	Method	KL_{AA} ($\downarrow$)	Novelty ($\uparrow$)	SeqID ($\downarrow$)
RRM	SA gen	$\mathbf{0.060 \pm 0.005}$	$\mathbf{0.513 \pm 0.011}$	$\mathbf{0.538 \pm 0.006}$
	SA ret	0.107 ± 0.010	0.334 ± 0.020	0.616 ± 0.012
	BS	0.133 ± 0.006	0.000 ± 0.000	0.645 ± 0.004
	GP	0.132 ± 0.005	0.008 ± 0.000	0.633 ± 0.006
	CC	2.091 ± 0.000	0.510 ± 0.009	0.562 ± 0.000
SH3	SA gen	0.036 ± 0.004	$\mathbf{0.471 \pm 0.012}$	$\mathbf{0.593 \pm 0.006}$
	SA ret	0.035 ± 0.004	0.213 ± 0.012	0.722 ± 0.011
	BS	$\mathbf{0.030 \pm 0.003}$	0.000 ± 0.000	0.772 ± 0.006
	GP	$\mathbf{0.030 \pm 0.003}$	0.006 ± 0.000	0.752 ± 0.006
	CC	0.798 ± 0.012	0.474 ± 0.008	0.601 ± 0.002
WW	SA gen	$\mathbf{0.008 \pm 0.002}$	$\mathbf{0.653 \pm 0.006}$	$\mathbf{0.562 \pm 0.005}$
	SA ret	0.046 ± 0.013	0.354 ± 0.016	0.763 ± 0.011
	BS	0.024 ± 0.003	0.000 ± 0.000	0.823 ± 0.005
	GP	0.029 ± 0.004	0.017 ± 0.000	0.809 ± 0.006
	CC	1.375 ± 0.092	0.637 ± 0.004	0.766 ± 0.001
Kunitz	SA gen	$\mathbf{0.013 \pm 0.002}$	$\mathbf{0.557 \pm 0.009}$	$\mathbf{0.607 \pm 0.004}$
	SA ret	0.037 ± 0.003	0.257 ± 0.016	0.760 ± 0.012
	BS	0.023 ± 0.002	0.000 ± 0.000	0.834 ± 0.004
	GP	0.024 ± 0.002	0.010 ± 0.000	0.844 ± 0.004
	CC	0.721 ± 0.013	0.546 ± 0.007	0.590 ± 0.001
zf-C2H2	SA gen	0.013 ± 0.028	$\mathbf{0.596 \pm 0.010}$	$\mathbf{0.557 \pm 0.007}$
	SA ret	0.021 ± 0.010	0.167 ± 0.014	0.848 ± 0.013
	BS	$\mathbf{0.007 \pm 0.002}$	0.000 ± 0.000	0.938 ± 0.004
	GP	$\mathbf{0.007 \pm 0.002}$	0.011 ± 0.000	0.940 ± 0.004
	CC	2.346 ± 0.060	0.580 ± 0.006	0.600 ± 0.001
PDZ	SA gen	$\mathbf{0.038 \pm 0.010}$	$\mathbf{0.418 \pm 0.009}$	$\mathbf{0.543 \pm 0.005}$
	SA ret	0.059 ± 0.016	0.236 ± 0.017	0.626 ± 0.014
	BS	0.041 ± 0.002	0.000 ± 0.000	0.642 ± 0.009
	GP	0.045 ± 0.003	0.006 ± 0.000	0.648 ± 0.007
	CC	0.339 ± 0.001	0.449 ± 0.009	0.549 ± 0.001
Pkinase	SA gen	$\mathbf{0.035 \pm 0.001}$	$\mathbf{0.478 \pm 0.011}$	0.515 ± 0.004
	SA ret	0.050 ± 0.001	0.316 ± 0.013	0.531 ± 0.006
	BS	0.054 ± 0.001	0.000 ± 0.000	0.522 ± 0.003
	GP	0.051 ± 0.001	0.005 ± 0.000	0.533 ± 0.003
	CC	0.062 ± 0.001	0.446 ± 0.009	$\mathbf{0.470 \pm 0.000}$
Defensin_beta	SA gen	$\mathbf{0.022 \pm 0.004}$	$\mathbf{0.402 \pm 0.013}$	$\mathbf{0.655 \pm 0.007}$
	SA ret	0.025 ± 0.008	0.166 ± 0.015	0.790 ± 0.015
	BS	0.020 ± 0.003	0.000 ± 0.000	0.836 ± 0.006
	GP	0.023 ± 0.003	0.006 ± 0.000	0.835 ± 0.006
	CC	0.708 ± 0.001	0.442 ± 0.009	0.716 ± 0.001

Table S2: Generation metrics for the three learned baselines (profile HMM emit, EvoDiff, and the MSA Transformer) across all eight Pfam families, computed identically to the SA generation analysis (150 sequences per family). The KL column reports amino acid composition KL divergence (↓ lower is better). Novelty: PCA-space cosine novelty (↑ higher is better). SeqID: nearest sequence identity to a stored pattern. Values are mean $\pm$ SE over 30 sub-blocks of the generated set. These are the per-family values shown in the main-text baseline comparison.

Family	Method	$\mathrm{KL}_{\mathrm{AA}}$ (↓)	Novelty (↑)	SeqID
RRM	HMM emit	0.009 ± 0.005	0.647 ± 0.005	0.209 ± 0.005
	EvoDiff	0.005 ± 0.006	0.593 ± 0.008	0.386 ± 0.005
	MSA Transformer	0.022 ± 0.003	0.572 ± 0.008	0.266 ± 0.007
SH3	HMM emit	0.009 ± 0.017	0.614 ± 0.004	0.290 ± 0.006
	EvoDiff	0.003 ± 0.021	0.434 ± 0.016	0.509 ± 0.011
	MSA Transformer	0.011 ± 0.016	0.544 ± 0.011	0.418 ± 0.008
WW	HMM emit	0.007 ± 0.020	0.706 ± 0.003	0.389 ± 0.009
	EvoDiff	0.009 ± 0.011	0.642 ± 0.006	0.509 ± 0.004
	MSA Transformer	0.043 ± 0.011	0.670 ± 0.006	0.417 ± 0.008
Kunitz	HMM emit	0.030 ± 0.011	0.665 ± 0.004	0.229 ± 0.006
	EvoDiff	0.011 ± 0.012	0.621 ± 0.008	0.434 ± 0.007
	MSA Transformer	0.024 ± 0.018	0.598 ± 0.009	0.338 ± 0.013
zf-C2H2	HMM emit	0.006 ± 0.038	0.670 ± 0.004	0.415 ± 0.008
	EvoDiff	0.003 ± 0.053	0.646 ± 0.007	0.469 ± 0.005
	MSA Transformer	0.014 ± 0.044	0.619 ± 0.007	0.477 ± 0.007
PDZ	HMM emit	0.012 ± 0.005	0.591 ± 0.004	0.143 ± 0.004
	EvoDiff	0.005 ± 0.007	0.344 ± 0.020	0.436 ± 0.016
	MSA Transformer	0.032 ± 0.004	0.367 ± 0.016	0.298 ± 0.015
Pkinase	HMM emit	0.019 ± 0.001	0.561 ± 0.006	0.112 ± 0.002
	EvoDiff	0.173 ± 0.025	0.554 ± 0.008	0.147 ± 0.005
	MSA Transformer	0.052 ± 0.004	0.535 ± 0.006	0.092 ± 0.002
Defensin_beta	HMM emit	0.031 ± 0.012	0.574 ± 0.007	0.362 ± 0.007
	EvoDiff	0.009 ± 0.035	0.389 ± 0.012	0.570 ± 0.009
	MSA Transformer	0.038 ± 0.052	0.316 ± 0.017	0.590 ± 0.015

Table S3: SA generation quality at $K{=}20$ across four Pfam families (five independent subsamples each). Values are mean $\pm$ SD over the five replicates.

Family	K_{full}	L	d_{PCA}	KL	Novelty	Seq. ID
SH3	55	48	17–18	0.026 ± 0.003	0.46 ± 0.01	0.67 ± 0.01
Kunitz	99	53	17–18	0.020 ± 0.004	0.47 ± 0.01	0.72 ± 0.01
zf-C2H2	151	23	18	0.013 ± 0.004	0.47 ± 0.01	0.73 ± 0.00
WW	420	31	17–18	0.019 ± 0.008	0.47 ± 0.01	0.71 ± 0.02

Table S4: Experimentally determined reference structures used for TM-align scoring.

Family	Pfam ID	PDB (chain)	Reference protein
RRM	PF00076	1FXL (A)	Sex-lethal RRM
SH3	PF00018	1SHG (A)	α-spectrin SH3 domain
WW	PF00397	1PIN (A)	Pin1 WW domain
Kunitz	PF00014	1BPI (A)	BPTI (Kunitz domain)
zf-C2H2	PF00096	1ZAA (C)	Zif268 zinc fingers
PDZ	PF00595	1BE9 (A)	PSD-95 PDZ3
Pkinase	PF00069	1ATP (E)	PKA catalytic subunit
Defensin_beta	PF00711	1E4S (A)	Beta-defensin

Table S5: Within-family Spearman correlation between a generated sequence's identity to the family consensus and its predicted structural quality (ESMFold pLDDT and reference-normalized TM-score), with the mean identity to consensus. $n=50$ generated sequences per family.

Family	ρ(consID, pLDDT)	ρ(consID, TM)	mean consID
RRM	0.21	0.36	0.60
SH3	0.30	-0.13	0.58
WW	0.29	-0.07	0.49
Kunitz	0.17	0.28	0.56
zf-C2H2	0.34	0.18	0.44
PDZ	0.43	0.53	0.56
Pkinase	-0.10	-0.09	0.73
Defensin_beta	0.28	0.36	0.58
Pooled (standardized)	0.23	0.17	n/a

Table S6: Pairwise mutual information (MI) preservation: Pearson correlation r between upper-triangle MI vectors of stored sequences and sequences generated by each method. Higher values indicate better preservation of pairwise residue covariation. *MSAT: MSA Transformer.

Family	SA gen	SA ret	Potts	HMM	EvoDiff	MSAT*
RRM	0.53	0.34	0.65	0.49	0.86	0.60
SH3	0.66	0.64	0.75	0.46	0.88	0.73
WW	0.71	0.49	0.88	0.29	0.83	0.58
Kunitz	0.80	0.69	0.89	0.32	0.87	0.50
zf-C2H2	0.92	0.87	0.96	0.72	0.97	0.92
PDZ	0.59	0.44	0.94	0.07	0.81	0.01
Pkinase	0.05	-0.09	0.95	0.23	0.59	0.05
Defensin_beta	0.85	0.75	0.23	0.83	0.95	0.88

Table S7: ESM2-650M pseudo-perplexity validation. Lower pseudo-perplexity indicates a more protein-like sequence. p: Wilcoxon rank-sum test (two-sided). d: Cohen's effect size (negative values indicate SA is better). Significance: $^{***}p < 0.001$, $^{**}p < 0.01$, n.s. = not significant ($p > 0.05$).

Family	SA gen ppl	Stored ppl	p	d	Sig.
RRM	4.87 ± 0.68	5.20 ± 1.92	0.30	-0.29	n.s.
SH3	5.24 ± 0.82	4.62 ± 1.48	< 0.001	0.60	***
WW	6.78 ± 1.77	5.04 ± 1.36	< 0.001	1.03	***
Kunitz	4.47 ± 0.57	4.98 ± 0.94	< 0.001	-0.74	***
zf-C2H2	7.26 ± 1.66	8.10 ± 3.47	0.41	-0.37	n.s.
PDZ	6.30 ± 1.21	5.31 ± 2.61	0.002	0.60	**
Pkinase	4.44 ± 0.32	4.32 ± 0.80	0.39	0.27	n.s.
Defensin_beta	6.35 ± 1.27	6.82 ± 2.09	0.32	-0.31	n.s.

Table S8: Deep mutational scanning validation. Substitutions generated by stochastic attention relative to the family consensus were matched to published single-mutant DMS fitness data. The tolerance rate is the fraction of matched substitutions classified as tolerated. The DMS overall column gives the baseline tolerance rate across all single mutations in the dataset. Enrichment is the ratio of the SA tolerance rate to the DMS overall rate.

Family	DMS source	Matched	SA tol. rate	DMS overall	Enrichment	p
PDZ	McLaughlin et al. 2012	4,313	91.7%	77.5%	$1.18\times$	$< 10^{-134}$
SH3	Faure et al. 2022	2,396	90.4%	67.5%	$1.34\times$	$< 10^{-155}$

Table S9: Position-matched and conservation-matched DMS nulls. The position-matched null fixes the set of positions SA substituted and asks how often a randomly chosen *tested* substitution at each such position is tolerated; the conservation-matched null draws comparison substitutions from positions in the same alignment-conservation quintile. Enrichment is the ratio of the observed SA tolerance rate to the null rate. The position-matched p is from a Poisson-binomial z-test (one-sided, SA $>$ null) and is corroborated by a $20{,}000$-draw permutation test ($p < 5 \times 10^{-5}$ in both families).

Family	SA tol. rate	Overall null	Position-matched null	Conservation-matched null	Pos.-matched p
PDZ	0.917 (n=4,313)	0.775 ($1.18\times$)	0.843 ($1.09\times$)	0.841 ($1.09\times$)	$< 10^{-77}$
SH3	0.904 (n=2,396)	0.675 ($1.34\times$)	0.749 ($1.21\times$)	0.744 ($1.22\times$)	$< 10^{-121}$

Table S10: Potts model (plmDCA) baseline comparison across all eight Pfam families. Metrics are computed identically to the main baseline analysis (30 chains $\times$ 5 samples). Standard errors are computed over 30 sub-blocks of five sequences (as in Table S2); consequently the KL SE for SA generation differs from the $1{,}000$-fold bootstrap KL SE reported in Table S1, although the SA-generation point estimates are identical between the two tables.

Family	Method	KL_{AA} ($\downarrow$)	Novelty ($\uparrow$)	SeqID
RRM	SA generation	0.060 ± 0.036	0.513 ± 0.011	0.538 ± 0.006
	Potts (plmDCA)	0.018 ± 0.015	0.562 ± 0.008	0.396 ± 0.011
SH3	SA generation	0.036 ± 0.026	0.471 ± 0.012	0.593 ± 0.007
	Potts (plmDCA)	0.025 ± 0.019	0.374 ± 0.011	0.539 ± 0.009
WW	SA generation	0.008 ± 0.012	0.653 ± 0.006	0.562 ± 0.005
	Potts (plmDCA)	0.011 ± 0.030	0.657 ± 0.007	0.436 ± 0.011
Kunitz	SA generation	0.013 ± 0.022	0.557 ± 0.009	0.607 ± 0.004
	Potts (plmDCA)	0.010 ± 0.017	0.540 ± 0.010	0.548 ± 0.010
zf-C2H2	SA generation	0.013 ± 0.058	0.596 ± 0.010	0.557 ± 0.007
	Potts (plmDCA)	0.005 ± 0.043	0.597 ± 0.009	0.512 ± 0.008
PDZ	SA generation	0.038 ± 0.017	0.418 ± 0.009	0.543 ± 0.005
	Potts (plmDCA)	0.012 ± 0.007	0.359 ± 0.022	0.302 ± 0.019
Pkinase	SA generation	0.035 ± 0.001	0.478 ± 0.011	0.515 ± 0.004
	Potts (plmDCA)	0.015 ± 0.001	0.222 ± 0.018	0.347 ± 0.022
Defensin_beta	SA generation	0.022 ± 0.058	0.402 ± 0.013	0.655 ± 0.007
	Potts (plmDCA)	0.112 ± 0.089	0.215 ± 0.011	0.753 ± 0.013

Table S11: Sampling diagnostics for ULA and MALA chains across eight Pfam families. The quantity τ_{int} is the integrated autocorrelation time of the Hopfield energy (mean over 10 chains). Effective sample size is reported per chain (n=3,000 post-burn-in samples); MALA AR is the Metropolis-adjusted acceptance rate; Conv. iter is the mean iteration at which burn-in energy converged (1% criterion); and Margin is the ratio of burn-in budget (2,000) to convergence iteration.

Family	$\tau_{\mathrm{int}}^{\mathrm{ULA}}$	ESS$^{\mathrm{ULA}}$	$\tau_{\mathrm{int}}^{\mathrm{MALA}}$	ESS$^{\mathrm{MALA}}$	MALA AR	Conv. iter	Margin
RRM	76	42	99	35	0.998	509	$3.9\times$
SH3	95	35	84	39	0.998	520	$3.8\times$
WW	86	37	96	34	0.996	369	$5.4\times$
Kunitz	100	35	109	33	0.997	500	$4.0\times$
zf-C2H2	105	33	82	39	0.997	447	$4.5\times$
PDZ	113	32	131	27	0.998	304	$6.6\times$
Pkinase	83	38	98	36	0.998	617	$3.2\times$
Defensin_beta	122	32	124	29	0.998	306	$6.5\times$

Table S12: Initialization sensitivity: stored-pattern vs. random sphere initialization. All metrics are computed using the same procedure as in the main analysis (30 chains $\times$ 5 samples). The two strategies produce similar output across all eight families.

Family	Init	KL$_{\mathrm{AA}}$	Novelty	SeqID	MaxCos
RRM	Stored	0.060	0.513	0.538	0.487
	Random	0.059	0.511	0.539	0.489
SH3	Stored	0.036	0.471	0.593	0.529
	Random	0.036	0.471	0.593	0.529
WW	Stored	0.008	0.653	0.562	0.347
	Random	0.008	0.653	0.562	0.347
Kunitz	Stored	0.013	0.557	0.607	0.443
	Random	0.013	0.557	0.607	0.443
zf-C2H2	Stored	0.013	0.596	0.557	0.404
	Random	0.013	0.596	0.557	0.404
PDZ	Stored	0.038	0.418	0.543	0.582
	Random	0.038	0.418	0.543	0.582
Pkinase	Stored	0.035	0.478	0.515	0.522
	Random	0.035	0.478	0.515	0.522
Defensin_beta	Stored	0.022	0.402	0.655	0.598
	Random	0.022	0.402	0.655	0.598

Table S13: Near-duplicate analysis: sequence identity between SA-generated and stored sequences. MaxID: mean maximum identity of each generated sequence to its nearest stored neighbor. %>95, %>90, %>80: fraction of generated sequences exceeding the indicated identity threshold. %≤2sub: fraction within 2 substitutions of a stored sequence. Stored ID: mean pairwise identity among stored sequences.

Family	Mean MaxID	%>95	%>90	%>80	%≤2sub	Stored ID
RRM	0.538	0.0	0.0	0.0	0.0	0.230
SH3	0.593	0.0	0.0	0.7	0.0	0.288
WW	0.562	0.0	0.0	0.0	0.0	0.324
Kunitz	0.607	0.0	0.0	0.0	0.0	0.368
zf-C2H2	0.557	0.0	0.0	0.0	0.0	0.304
PDZ	0.543	0.0	0.0	0.7	0.0	0.217
Pkinase	0.515	0.0	0.0	0.0	0.0	0.274
Defensin_beta	0.655	0.0	0.0	6.0	0.0	0.332

Table S14: Empirical within-family nearest-neighbor identity envelope. For each family, $\bar{I}_{\mathrm{NN}}$ is the mean over stored sequences of each sequence's leave-one-out nearest-neighbor identity (identity to its closest neighbor in the same family), with standard deviation, 10th/90th percentiles, and range. "SA SeqID" is the mean nearest-neighbor identity of generated sequences (from Table S1), and "Gen. pair ID" is the mean pairwise identity among the 150 generated sequences. All 150 generated sequences were distinct in every family.

Family	$\bar{I}_{\mathrm{NN}}$	SD	p_{10}	p_{90}	[min, max]	SA SeqID	Gen. pair ID
RRM	0.397	0.099	0.292	0.483	$[0.254, 0.803]$	0.538	0.483
SH3	0.506	0.099	0.396	0.625	$[0.292, 0.708]$	0.593	0.476
WW	0.601	0.074	0.516	0.677	$[0.387, 0.806]$	0.562	0.376
Kunitz	0.568	0.117	0.449	0.736	$[0.358, 0.792]$	0.607	0.477
zf-C2H2	0.479	0.048	0.435	0.522	$[0.304, 0.565]$	0.557	0.347
PDZ	0.526	0.179	0.301	0.711	$[0.217, 0.807]$	0.543	0.445
Pkinase	0.381	0.046	0.325	0.431	$[0.275, 0.454]$	0.515	0.645
Defensin_beta	0.584	0.122	0.400	0.722	$[0.333, 0.778]$	0.655	0.470

Table S15: Wall-clock time for generating 150 sequences per family. Stochastic attention and HMM times are averaged over three runs on a single CPU core (Apple M2). EvoDiff and MSA Transformer times are from CPU-only runs. Potts times include pseudo-likelihood fitting and Gibbs sampling.

Family	L	K	SA (s)	HMM (s)	EvoDiff (h)	MSAT (h)	Potts (min)
RRM	71	68	0.5	0.05	~ 3	~ 4	~ 4
SH3	48	55	0.4	0.03	~ 2	~ 2	~ 2
WW	31	420	4.6	0.03	~ 2	~ 3	~ 1
Kunitz	53	99	0.7	0.03	~ 2	~ 3	~ 2
zf-C2H2	23	151	1.0	0.02	~ 2	~ 1	~ 1
PDZ	83	44	0.3	0.04	~ 4	~ 4	~ 5
Pkinase	262	37	0.2	0.09	~ 14	~ 8	~ 52
Defensin_beta	36	45	0.3	0.03	~ 2	~ 1	~ 1

Table S16: Consensus-with-noise baseline vs. SA generation. The label CN denotes the consensus sequence with calibrated Gaussian noise in PCA space; SA denotes stochastic attention generation ($\beta \approx 2\beta^*$). Metrics are computed identically to the main analysis.

Family	Method	$\mathrm{KL}_{\mathrm{AA}}$	Novelty	SeqID
RRM	CN	0.013 ± 0.002	0.686 ± 0.005	0.487 ± 0.005
	SA gen	0.060 ± 0.005	0.513 ± 0.011	0.538 ± 0.006
SH3	CN	0.004 ± 0.001	0.658 ± 0.005	0.537 ± 0.005
	SA gen	0.036 ± 0.004	0.471 ± 0.012	0.593 ± 0.006
WW	CN	0.021 ± 0.003	0.787 ± 0.002	0.437 ± 0.003
	SA gen	0.008 ± 0.004	0.653 ± 0.006	0.562 ± 0.005
Kunitz	CN	0.004 ± 0.001	0.722 ± 0.004	0.551 ± 0.003
	SA gen	0.013 ± 0.002	0.557 ± 0.009	0.607 ± 0.004
zf-C2H2	CN	0.095 ± 0.002	0.742 ± 0.003	0.495 ± 0.004
	SA gen	0.013 ± 0.027	0.596 ± 0.010	0.557 ± 0.007
PDZ	CN	0.021 ± 0.008	0.633 ± 0.005	0.522 ± 0.005
	SA gen	0.038 ± 0.010	0.418 ± 0.009	0.543 ± 0.005
Pkinase	CN	0.015 ± 0.001	0.630 ± 0.005	0.528 ± 0.004
	SA gen	0.035 ± 0.001	0.478 ± 0.011	0.515 ± 0.004
Defensin_beta	CN	0.013 ± 0.002	0.631 ± 0.005	0.620 ± 0.006
	SA gen	0.022 ± 0.003	0.402 ± 0.013	0.655 ± 0.007

Table S17: Column-permuted alignment control. The label SA (real) denotes stochastic attention on the original alignment, and SA (perm) denotes stochastic attention on a column-permuted alignment that preserves per-position frequencies but destroys inter-position correlations; MI r is the Pearson correlation between pairwise MI vectors of stored and generated sequences.

Family	Method	$\mathrm{KL}_{\mathrm{AA}}$	Novelty	SeqID	MI r
RRM	SA (real)	0.060 ± 0.005	0.513 ± 0.011	0.538 ± 0.006	0.53
	SA (perm)	0.050 ± 0.017	0.629 ± 0.005	0.460 ± 0.003	0.48
SH3	SA (real)	0.036 ± 0.004	0.471 ± 0.012	0.593 ± 0.006	0.67
	SA (perm)	0.032 ± 0.004	0.611 ± 0.005	0.500 ± 0.003	0.59
WW	SA (real)	0.008 ± 0.002	0.653 ± 0.006	0.562 ± 0.005	0.73
	SA (perm)	0.006 ± 0.002	0.688 ± 0.004	0.540 ± 0.004	0.73
Kunitz	SA (real)	0.013 ± 0.002	0.557 ± 0.009	0.607 ± 0.004	0.80
	SA (perm)	0.012 ± 0.002	0.660 ± 0.007	0.546 ± 0.003	0.78
zf-C2H2	SA (real)	0.013 ± 0.027	0.596 ± 0.010	0.557 ± 0.007	0.92
	SA (perm)	0.006 ± 0.005	0.654 ± 0.006	0.511 ± 0.004	0.93
PDZ	SA (real)	0.038 ± 0.011	0.418 ± 0.009	0.543 ± 0.005	0.58
	SA (perm)	0.028 ± 0.005	0.545 ± 0.007	0.433 ± 0.003	0.52
Pkinase	SA (real)	0.035 ± 0.001	0.478 ± 0.011	0.515 ± 0.004	0.06
	SA (perm)	0.035 ± 0.001	0.612 ± 0.005	0.449 ± 0.001	0.00
Defensin_beta	SA (real)	0.022 ± 0.003	0.402 ± 0.013	0.655 ± 0.007	0.85
	SA (perm)	0.017 ± 0.002	0.542 ± 0.008	0.552 ± 0.004	0.82

Table S18: PCA variance threshold sensitivity study. For each family and retained-variance threshold, the table reports the PCA dimension d, empirical critical temperature β^*, generation temperature β_{gen}, amino acid composition KL divergence, PCA-space cosine novelty, and nearest sequence identity. The horizontal rule within each family separates the degraded regime ($\leq 70\%$) from the robust regime ($\geq 80\%$).

Family	Threshold	d	β^*	β_{gen}	$\mathrm{KL}_{\mathrm{AA}}$	Novelty	SeqID
RRM	50%	21	3.85	8	0.186	0.392	0.539
	60%	27	3.85	8	0.111	0.426	0.546
	70%	34	3.85	8	0.135	0.463	0.539
	80%	43	3.85	8	0.085	0.497	0.535
	90%	53	3.85	8	0.067	0.508	0.540
	95%	59	3.85	8	0.060	0.513	0.538
SH3	50%	15	3.23	6	0.087	0.312	0.601
	60%	19	3.23	6	0.072	0.372	0.593
	70%	25	3.23	6	0.039	0.402	0.592
	80%	32	3.23	6	0.038	0.447	0.587
	90%	41	3.85	8	0.032	0.444	0.596
	95%	46	3.85	8	0.036	0.471	0.593
WW	50%	33	5.45	11	0.152	0.410	0.646
	60%	47	5.45	11	0.133	0.476	0.631
	70%	67	5.45	11	0.125	0.530	0.624
	80%	95	5.45	11	0.028	0.588	0.600
	90%	142	5.45	11	0.020	0.636	0.576
	95%	186	5.45	11	0.008	0.653	0.562
Kunitz	50%	23	3.85	8	0.053	0.387	0.638
	60%	31	3.85	8	0.047	0.433	0.636
	70%	40	3.85	8	0.031	0.469	0.628
	80%	53	3.85	8	0.022	0.500	0.619
	90%	69	3.85	8	0.023	0.535	0.616
	95%	80	3.85	8	0.013	0.557	0.607
zf-C2H2	50%	28	4.58	9	0.112	0.397	0.610
	60%	38	4.58	9	0.037	0.458	0.599
	70%	49	4.58	9	0.106	0.483	0.595
	80%	65	4.58	9	0.017	0.528	0.583
	90%	88	4.58	9	0.010	0.565	0.572
	95%	106	4.58	9	0.013	0.596	0.557
PDZ	50%	12	2.70	5	0.038	0.269	0.558
	60%	16	3.23	6	0.035	0.305	0.564
	70%	20	3.23	6	0.042	0.337	0.553
	80%	25	3.23	6	0.035	0.359	0.549
	90%	32	3.23	6	0.046	0.400	0.544
	95%	37	3.23	6	0.038	0.418	0.543
Pkinase	50%	15	3.23	6	0.041	0.356	0.501
	60%	18	3.23	6	0.041	0.390	0.500
	70%	22	3.23	6	0.038	0.426	0.506
	80%	27	3.23	6	0.035	0.463	0.506
	90%	31	3.23	6	0.035	0.480	0.509
	95%	34	3.23	6	0.035	0.478	0.515
Defensin_beta	50%	11	2.70	5	0.036	0.223	0.679
	60%	15	3.23	6	0.035	0.262	0.678
	70%	19	3.23	6	0.030	0.307	0.668
	80%	25	3.23	6	0.029	0.312	0.680
	90%	32	3.23	6	0.020	0.381	0.657
	95%	37	3.23	6	0.022	0.402	0.655

Table S19: Leave-one-family-out cross-validation of β^* prediction. Scheme 1: full 33-point dataset. Scheme 2: 8-family-only LOO. Both schemes support out-of-sample generalization.

Scheme	n	LOOCV R^2	LOOCV RMSE
Full 33-pt LOFO	33	0.959	0.184
8-family LOO	8	0.928	0.195

Table S20: Summary statistics for Pfam 36.0 family sizes. Values report the number of sequences per family across all Pfam-A entries.

Statistic	Seed alignments	Full alignments
Total families	20,795	20,652
Median	22	841
Mean	59.5	5,075.9
25th percentile	7	127
75th percentile	65	2,973
Families < 20	9,888 (47.5%)	2,028 (9.8%)
Families < 50	14,386 (69.2%)	3,438 (16.6%)
Families < 100	17,523 (84.3%)	4,684 (22.7%)

Table S21: Fixed random seeds by experiment.

Experiment	Implementation	Seed(s)
Main sequence generation	Julia	42 (setup); $12345{+}c$ per chain
HMM validation	Julia	42; $12345{+}c$ per chain
Sampling diagnostics (ULA/MALA)	Julia	42; $12345{+}c$ per chain
Multifamily scaling	Julia	per-replicate seed
Baseline evaluation and bootstrap	Julia	12345
Permuted-alignment control	Julia	54321; 12345
PCA sensitivity	Julia	42; per-condition seed
$K{=}20$ specificity	Julia	20001
Potts (plmDCA) baseline	Python	42
MSA Transformer baseline	Python	42
AlphaFold2 via ColabFold	Python	0
DMS position-matched null	Python	0